

MedTrack_DV

Hospital Operations & Patient Analytics Dashboard

MODULE 8 – DOCUMENTATION & PROJECT DELIVERY

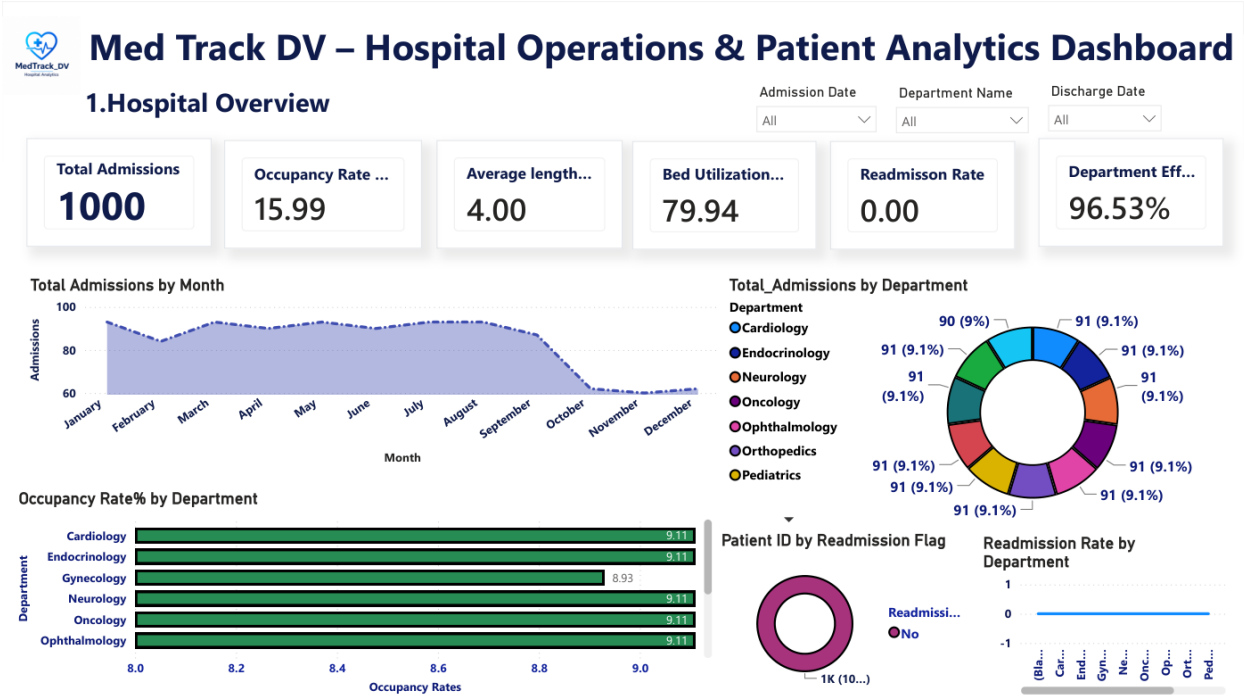


Figure 1. Submitted Hospital Overview dashboard snapshot.

Documentation objective

This document consolidates the data source register, KPI definitions, calculation methodology, dashboard operating guide, healthcare operations methodology, validation notes, project structure, governance controls, and delivery checklist for the MedTrack_DV project.

Document	Module 8 – Documentation & Project Delivery
Project	MedTrack_DV – Hospital Operations & Patient Analytics Dashboard
Release	Version 1.0 17 August 2026
Evidence base	Supplied dataset, KPI script, equipment workbook, dashboard export and project specification

Prepared as a portfolio-ready project document based on the supplied project specification, dashboard exports, KPI-generation script, hospital dataset, and equipment-utilization workbook.

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Page numbers refer to the final PDF layout and have been adjusted to account for this index page.

1. Executive Summary

Document design: The report is structured so that a reviewer can move from requirements to evidence, from evidence to calculation logic, and from calculation logic to dashboard interpretation without needing to reconstruct the project from separate files. It also records assumptions and limitations explicitly so that the document remains credible during technical or academic review.

Module 8 closes the project lifecycle by converting the technical work into a traceable, review-ready deliverable.

MedTrack_DV is an analytics solution designed to convert hospital operational records into management information. The supplied project specification describes a unified dashboard suite covering Hospital Overview, Patient Flow, Department Analytics, and Resource Utilization, with the purpose of supporting patient-flow monitoring, department performance evaluation, resource optimization, and data-driven operational decisions.

The submitted implementation contains 1,000 patient/appointment-level records spanning 2023–2025, 11 departments, treatment-cost information, length-of-stay measures, bed fields, staff counts, admission/discharge dates, readmission flags, and equipment-utilization data. The supplied Hospital Overview dashboard reports 1,000 total admissions, 4.00 average length of stay, 0.00% readmission rate, 15.99% occupancy rate, 79.94% bed utilization, and 96.53% department efficiency.

Project Dimension	Documented Outcome
Data foundation	1,000-row operational dataset; 56 columns
KPI layer	6 core hospital KPIs
Dashboard layer	4 interconnected operational views
Resource layer	Beds, staff, capacity and equipment
QA layer	KPI, interaction and patient-flow validation
Delivery layer	Documentation, structure and deployment readiness

Professional positioning

The report is intentionally written as a controlled project artifact: every major metric is tied to a source, calculation rule, dashboard use case, or validation decision.

Management interpretation: The executive view should communicate the scale, purpose and status of the solution before the reviewer examines individual charts. The strongest message is that MedTrack_DV connects operational volume, patient flow, departmental performance and resource utilization in one governed analytics workflow. The documentation therefore treats the dashboard as a management-information product rather than a collection of independent charts.

2. Project Objectives, Scope and Boundaries

The project specification defines a complete lifecycle from data collection through documentation and delivery. Module 8 therefore does not introduce a new analytics model; it documents the existing implementation in a way that makes the solution understandable, reproducible, auditable, and portfolio-ready.

- Document the dataset sources and the role of each supplied data asset.
- Define each KPI in business language and calculation language.
- Explain how the four dashboard areas support hospital operations.
- Describe the healthcare operations methodology used to interpret patient flow, capacity, staffing, and equipment.
- Record data-processing assumptions and calculation dependencies.
- Provide a user guide for filters, navigation, KPI interpretation, and dashboard review.
- Create a professional project structure covering scripts, data, dashboard and documentation.
- Capture quality controls, limitations, risks, and recommended future improvements.

In Scope	Out of Scope / Controlled Boundary
Operational analytics	Clinical diagnosis or treatment recommendation
Admissions and discharges	Individual medical advice
Department performance	Patient-level clinical decision-making
Bed/staff/equipment utilization	Real-time hospital command-and-control
Revenue as operational indicator	Financial accounting or statutory reporting
Dashboard interpretation	Replacement of hospital information systems

Governance principle

The dashboard should be treated as an operational analytics layer, not as a clinical decision-support system.

Review focus: The project boundary is intentionally operational. The solution measures activity, capacity and resource patterns, but it does not replace clinical judgement, hospital information systems or audited financial reporting. Keeping this boundary explicit protects the credibility of the project and makes the intended audience clear.

3. Dataset Source Register

The supplied project specification states that the solution uses publicly available Hospital Datasets and Patient Admission Datasets and that the workflow includes collection, integration, cleaning and transformation. The exact public provider names or URLs are not specified in the supplied specification. For that reason, this documentation does not invent a provider or external URL.

Asset	Role in Project	Evidence / Scope
hospital_final_dataset CSV	Primary analytical dataset	1,000 rows; 56 columns; admissions, patients, departments, costs, beds and staffing
Equipment_Utilization XLSX	Equipment resource layer	8 equipment records with department and utilization percentage
generate_hospital_kpis.py	KPI calculation engine	Validation, cleaning, derived fields, KPI calculations and exports
Dashboard PDF	Presentation evidence	Four submitted dashboard pages: Overview, Patient Flow, Department Analytics, Resource Utilization
Project specification PDF	Requirements baseline	Module plan, KPI list, dashboard scope and Module 8 deliverables

The working dataset contains fields for patient identity, appointment information, treatment, department, doctor, dates, costs, bed assignment, bed status, readmission flag, staffing, and precomputed KPI fields. The KPI engine also maps source-column variants into a standardized analytical schema.

Source transparency

For an academic/portfolio submission, retain the original source citation or URL in the repository README if the external dataset provider is known. The current supplied project materials identify the source category but not the provider.

Evidence note: The source register distinguishes between the project requirement statement and the actual working files supplied for implementation. This is important because the supplied project specification describes public hospital and patient-admission datasets without naming a specific provider. The documentation records that limitation instead of inventing a citation.

4. Dataset Profile and Data Dictionary Overview

The working dataset contains 1,000 records and 56 columns. It combines appointment-level information with patient, doctor, department, treatment, billing, admission, discharge, bed, staffing, and KPI-related fields. This structure supports a broad operational dashboard without requiring multiple analytical tables for the submitted snapshot.

Domain	Representative Fields	Analytical Purpose
Patient	Patient Id, First Name, Last Name, Gender, Date Of Birth	Patient segmentation and identity
Appointment	Appointment Id, Date, Time, Status, Reason For Visit	Access and scheduling analysis
Clinical operations	Admission Date, Discharge Date, Length of Stay	Patient-flow and stay analysis
Department	Department Name, Head Of Department	Department comparisons
Provider	Doctor Id, Doctor Name, Specialization, Years Experience	Staff/provider context
Finance	Cost, Amount, Payment Status, Insurance Provider	Revenue and operational cost context
Beds	Bed ID, Total Beds, Bed Status	Capacity and utilization
Quality	Readmission Flag	Readmission monitoring
Resources	Staff Count, equipment workbook	Resource allocation and utilization

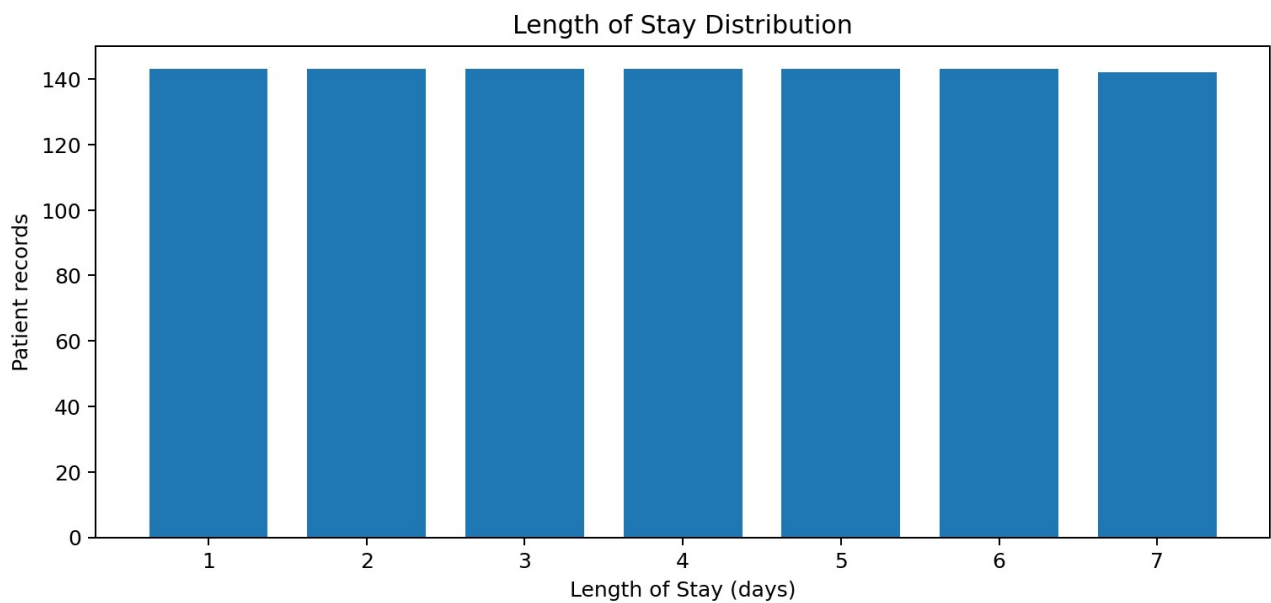


Figure 2. Distribution of recorded length of stay values.

Review focus: The data dictionary is designed around analytical domains rather than individual columns alone. This makes the dataset easier to understand during dashboard maintenance because each field can be associated with a business purpose such as patient flow, department comparison, resource planning or operational finance.

5. Data Quality Assessment and Preparation

The supplied KPI engine implements a controlled preparation sequence: load the CSV, rename source fields into the KPI-engine schema, construct patient and doctor names where required, derive patient age from date of birth, validate required columns, convert date and numeric fields, standardize department and categorical text, and create derived time fields.

Quality Check	Implementation / Observation
Row count	1,000 records loaded
Column count	56 source columns
Department coverage	11 department names
Admission period	01-01-2023 to 31-12-2025
Length of stay	1–7 days in supplied records
Readmission flag	All supplied records marked No
Bed status	All supplied records marked Occupied
Total beds field	200 in supplied records
Patient identifier reuse	48 unique Patient Id values across 1,000 rows

Important data-governance note

The dataset is an operational analytics working dataset. Patient names, phone numbers, addresses, emails and insurance identifiers appear in the source. A production deployment must apply de-identification, access control and minimum-necessary data principles before public publication.

Governance note: Healthcare analytics should treat data quality as part of the measurement process. Identifier reuse, categorical consistency, missing values, date validity and privacy exposure can all change the meaning of a KPI. The preparation layer therefore needs to be repeatable and documented rather than performed manually inside a dashboard.

6. Data Preparation Methodology

1. Load the supplied hospital_final_dataset.csv and verify file availability.
2. Rename source fields such as Patient ID, Department Name, Admission Date, Discharge Date, Length of Stay and Cost into the KPI-engine naming convention.
3. Construct Patient_Name and Doctor_Name from component name fields when needed.
4. Derive Patient_Age from Date Of Birth using the current date during execution.
5. Apply default values for missing Gender, Admission Type and Discharge Status where the engine requires a value.
6. Validate the required analytical columns before KPI calculation.
7. Convert admission/discharge dates to datetime and treatment cost, length of stay and patient age to numeric values.
8. Standardize Department, Gender, Admission Type and Discharge Status text formats.
9. Create Admission_Month, Admission_Year and Admission_Day derived fields.
10. Generate hospital, department, monthly and resource summary outputs and an Excel report.

This sequence separates preparation from measurement. That separation is important because a KPI should not be calculated from uncontrolled raw types or inconsistent category labels.

Reproducibility

The supplied Python engine is modular: loading, validation, cleaning, derived-column preparation, KPI calculation, summary generation and export are separated into functions. This supports repeatable refreshes.

Reproducibility note: The supplied KPI engine is structured as a sequence of functions, which makes the transformation pipeline easier to test and maintain. A future refresh should use the same order: load, validate, clean, derive, calculate, summarize and export. This provides a stable audit trail between the source file and the dashboard.

7. KPI Framework – Six Core Measures

The project specification requires six hospital KPIs: Total Admissions, Occupancy Rate, Average Length of Stay, Readmission Rate, Bed Utilization Rate, and Department Efficiency Score. These measures collectively cover demand, capacity, patient stay, quality signal, resource utilization, and departmental performance.

KPI	Business Question	Primary Dimension
Total Admissions	How much patient activity occurred?	Volume
Occupancy Rate	How much of fixed bed capacity was occupied?	Capacity
Average Length of Stay	How long did patients remain admitted?	Flow
Readmission Rate	How frequently were records marked as readmitted?	Quality
Bed Utilization Rate	How intensively was calculated bed capacity used?	Resource
Department Efficiency Score	Which departments combine volume, revenue and LOS performance effectively?	Efficiency

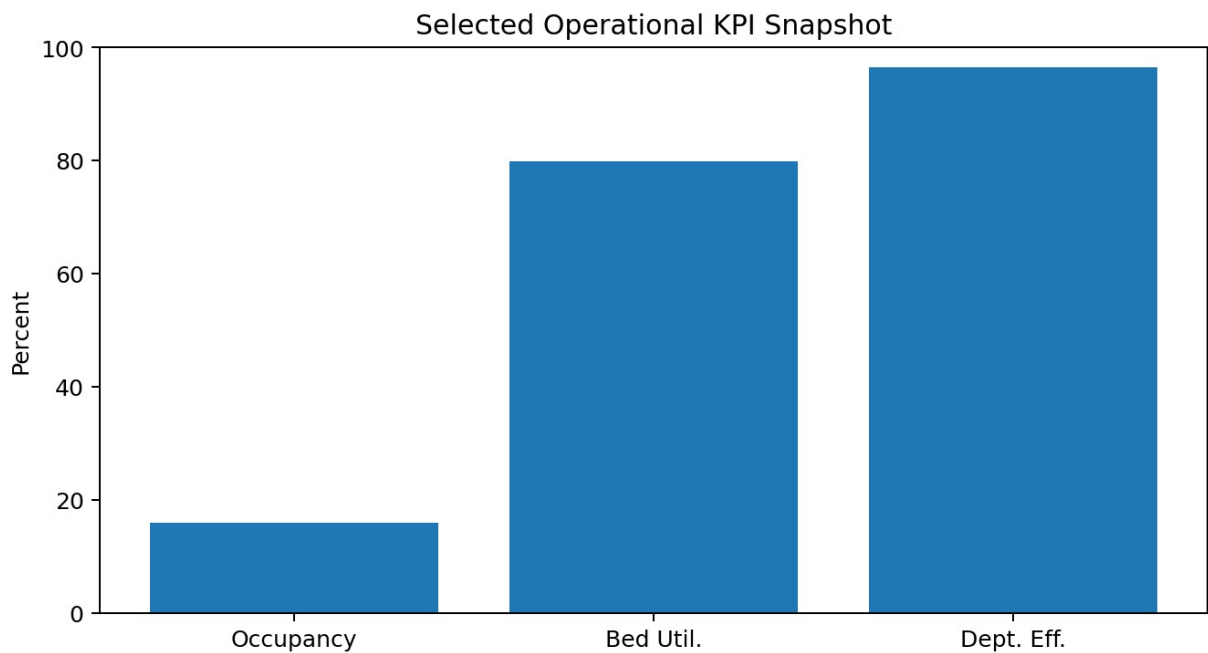


Figure 3. Selected KPI snapshot aligned with the submitted Hospital Overview dashboard.

Management interpretation: The six KPIs are complementary. Admissions describes demand, ALOS describes duration, occupancy and utilization describe capacity pressure, readmission provides a quality-related signal, and efficiency provides a comparative departmental index. Reading them together is more informative than optimizing any one measure in isolation.

8. KPI Definitions – Volume and Patient Flow

Total Admissions. Definition: the number of analytical records in the dataset. The supplied KPI engine implements this as the row count of the prepared dataframe. The submitted dashboard reports 1,000.

Average Length of Stay (ALOS). Definition: the arithmetic mean of Length_of_Stay across the analytical records. The supplied engine rounds the result to two decimal places. The submitted dashboard reports 4.00 days.

Measure	Formula / Rule	Dashboard Use
Total Admissions	count(records)	Headline volume KPI
ALOS	sum(LOS) / count(records)	Stay-duration card and department comparison
Monthly Admissions	count(records) grouped by admission month	Demand trend
Department Volume	count(records) grouped by department	Service-line comparison

Interpretation: admissions measure demand, while ALOS adds the time dimension. A high admission count with a high ALOS can imply sustained bed pressure; a lower admission count with a high ALOS can also consume substantial capacity.

Operational caution

ALOS is an operational measure, not a quality score by itself. It must be interpreted with case mix, clinical pathways, discharge readiness and resource constraints.

Review focus: Volume and flow measures should always be interpreted with their time context. A monthly count is not equivalent to a full-period total, and ALOS can change even when admissions remain stable. Dashboard filters should therefore be treated as part of the metric definition.

9. KPI Definitions – Capacity and Quality

Occupancy Rate. The KPI engine calculates occupied bed-days from total length of stay, divides by assumed total beds multiplied by the number of days in the admission period, and caps the result at 100%. The supplied implementation assumes 25 beds for this calculation.

Bed Utilization Rate. The KPI engine derives a bed denominator using occupied bed-days, an 80% utilization reference and the number of days in the observed period, then calculates occupied bed-days divided by calculated available bed-days.

Readmission Rate. If a Readmission field exists, the engine counts records with Yes/True/1 and divides by total records. Otherwise, it uses duplicate Patient_ID records as a proxy. In the supplied dataset, Readmission Flag is 'No' for all 1,000 records, while the submitted dashboard reports 0.00%.

KPI	Interpretation	Management Use
Occupancy	Capacity pressure indicator	Bed planning
Bed Utilization	Intensity of bed use	Resource planning
Readmission	Quality/return signal	Follow-up review

Definition discipline

Occupancy and bed utilization are related but should not be treated as interchangeable. The documentation preserves the calculation logic implemented in the supplied KPI engine.

Governance note: Capacity and quality metrics require especially careful definitions. Occupancy depends on the bed-capacity assumption, while readmission depends on the availability and reliability of longitudinal patient records. The documentation preserves the supplied implementation logic and identifies assumptions that should be reviewed before production use.

10. Department Efficiency Score – Methodology

The Department Efficiency Score is a composite operational indicator. The supplied KPI engine groups records by department and calculates admissions, revenue and average LOS. Each component is normalized relative to the best observed department, then combined with explicit weights.

Component	Weight	Normalization
Admissions score	40%	Department admissions / maximum department admissions
Revenue score	40%	Department revenue / maximum department revenue
LOS score	20%	Minimum department ALOS / department ALOS
Final score	100%	Weighted components $\times$ 100

The score therefore rewards higher relative volume and revenue while rewarding lower relative average length of stay. The resulting average efficiency in the supplied calculation is 96.53, matching the submitted dashboard headline.

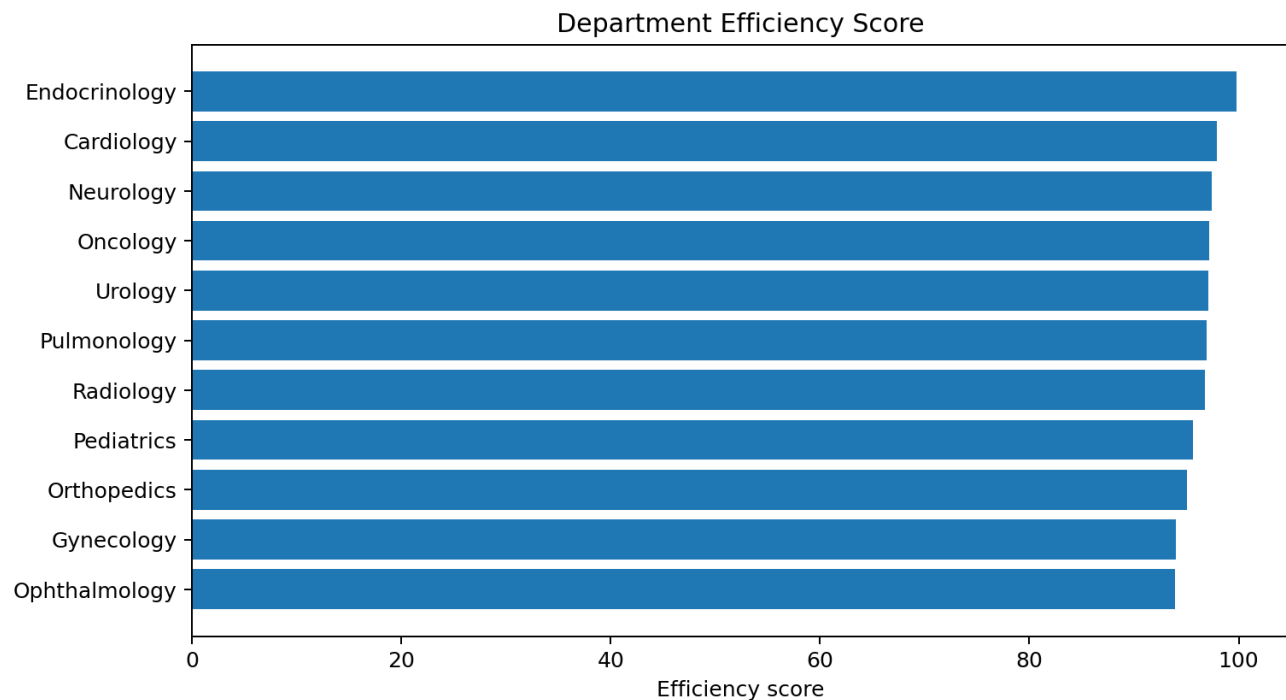


Figure 4. Department Efficiency Score calculated using the supplied KPI-engine methodology.

Interpretation

Efficiency is a comparative operational index, not a clinical outcome. A department should not be ranked solely by this score without reviewing workload, case complexity, staffing, equipment and patient outcomes.

Management interpretation: The composite efficiency score is useful because it balances three different operational dimensions. However, it remains a management index rather than a clinical quality measure. A responsible review should examine the underlying admissions, revenue and LOS components before acting on a departmental ranking.

11. Dashboard Guide – Architecture and Navigation

The supplied project specification defines four integrated dashboards: Hospital Overview, Patient Flow, Department Analytics and Resource Utilization. The submitted dashboard export is presented in Power BI Desktop, while the original project specification identifies Tableau as the planned visualization technology. This document treats the supplied Power BI export as the implementation evidence for the four-view dashboard specification.

Dashboard	Primary User Question	Key Visuals
Hospital Overview	What is the overall operational position?	KPI cards, monthly admissions, department volume, occupancy, readmission
Patient Flow	How are admissions, discharges and stay patterns moving?	Admission/discharge trends, movement view, ALOS, monthly patient counts
Department Analytics	Which departments drive volume, revenue and efficiency?	Department volume, efficiency, revenue, treatment capacity
Resource Utilization	How are beds, staff and equipment being used?	Bed utilization, occupied bed-days, staff, capacity, equipment

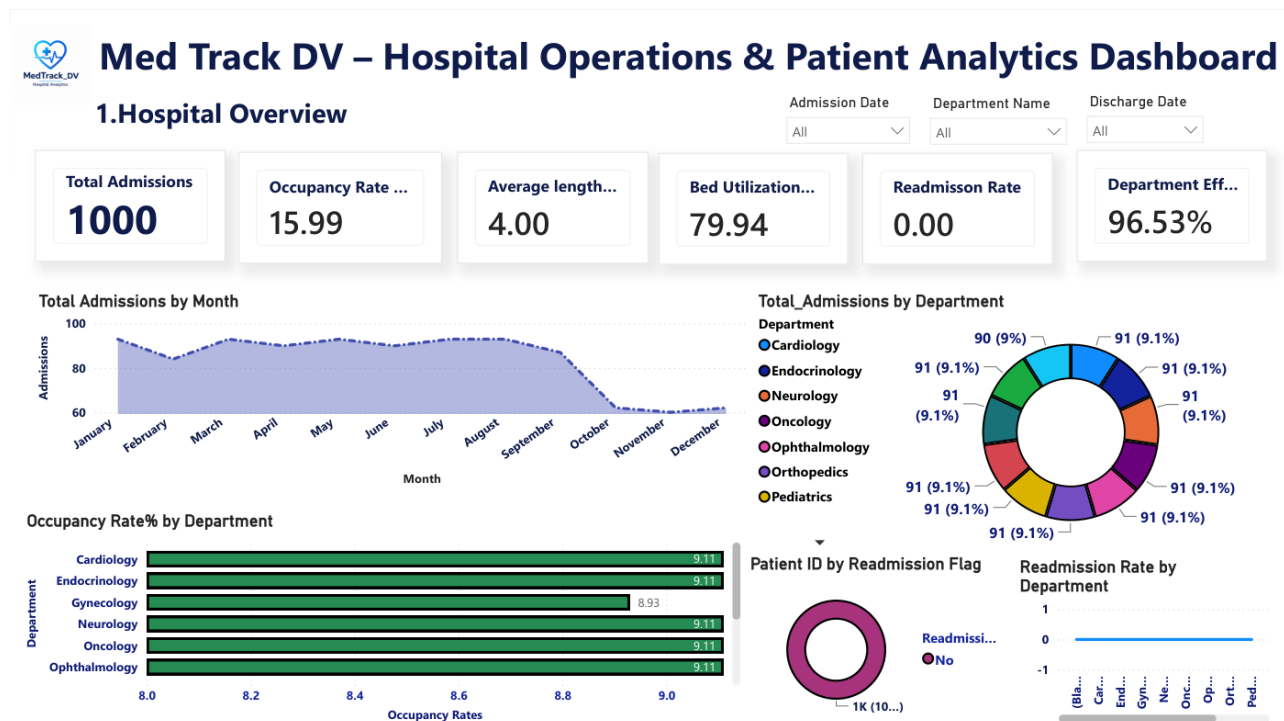


Figure 5. Hospital Overview dashboard.

Navigation note: The dashboard architecture follows the user journey from hospital-wide status to patient flow, then department performance and finally resources. This progression reduces the need for users to jump between unrelated visuals and supports a consistent management review sequence.

12. Dashboard Guide – Hospital Overview

The Hospital Overview page is the executive entry point. It exposes admission date, department and discharge date filters and presents the six core operational indicators in a compact KPI band.

- Use Admission Date to restrict the reporting period.
- Use Department Name to isolate one service line or compare departments.
- Use Discharge Date to focus the view on discharge activity.
- Read the KPI cards first to establish the current operational baseline.
- Use the monthly admissions chart to identify demand variation.
- Use department volume and occupancy visuals to locate service-line concentration.
- Review readmission visuals as a quality-monitoring signal, not a clinical conclusion.

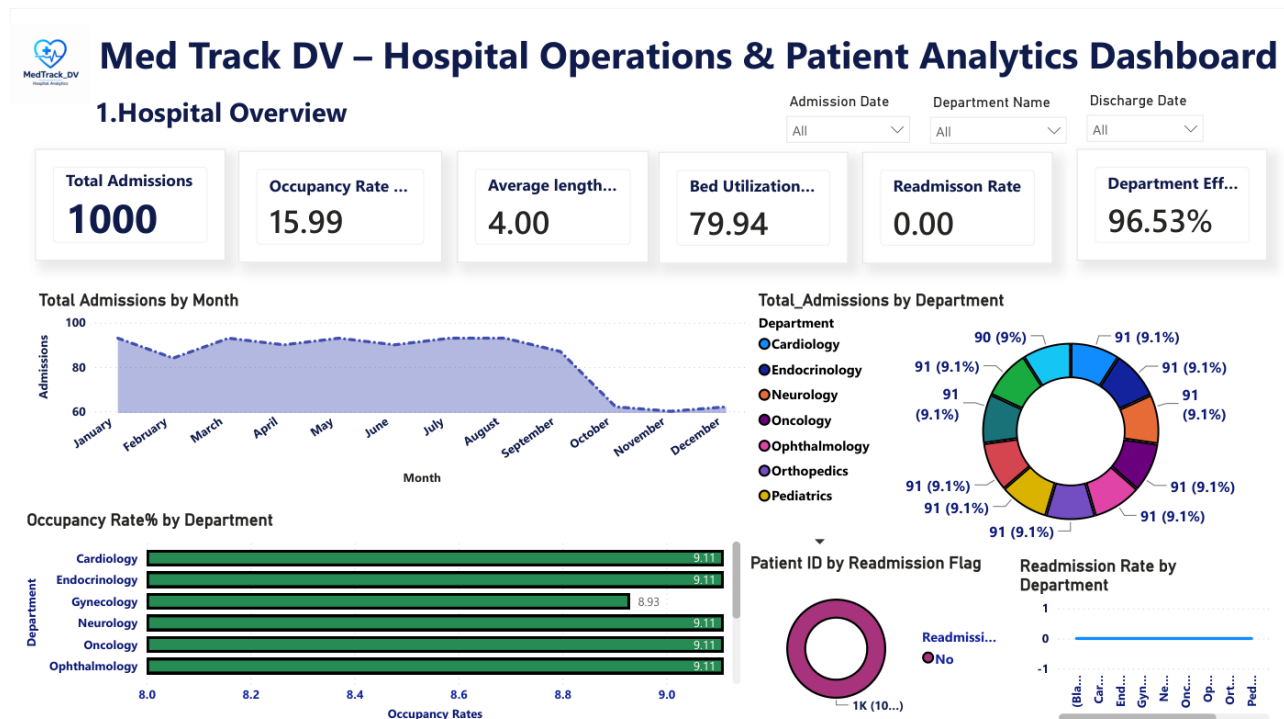


Figure 6. Hospital Overview – submitted dashboard evidence.

Expected baseline

The supplied snapshot shows 1,000 admissions, 4.00 average LOS, 0.00 readmission rate, 15.99 occupancy rate, 79.94 bed utilization and 96.53% department efficiency.

User guidance: Start with the KPI cards, confirm the filter context, then move from the monthly trend to department comparisons and readmission visuals. The overview is most useful when it answers three questions quickly: how much activity occurred, how much capacity is being consumed, and where should management investigate further?

13. Dashboard Guide – Patient Flow

Patient Flow focuses on the movement of patients through admission and discharge processes. It provides a monthly view of admissions and discharges, a patient movement perspective by year and department, average LOS by department, and monthly patient counts.

- Compare admission and discharge patterns for potential flow imbalance.
- Use ALOS by department to identify service lines with longer stays.
- Review monthly patient counts to locate high-load periods.
- Use year and department selections to narrow movement analysis.
- Interpret peaks as operational demand signals requiring contextual investigation.

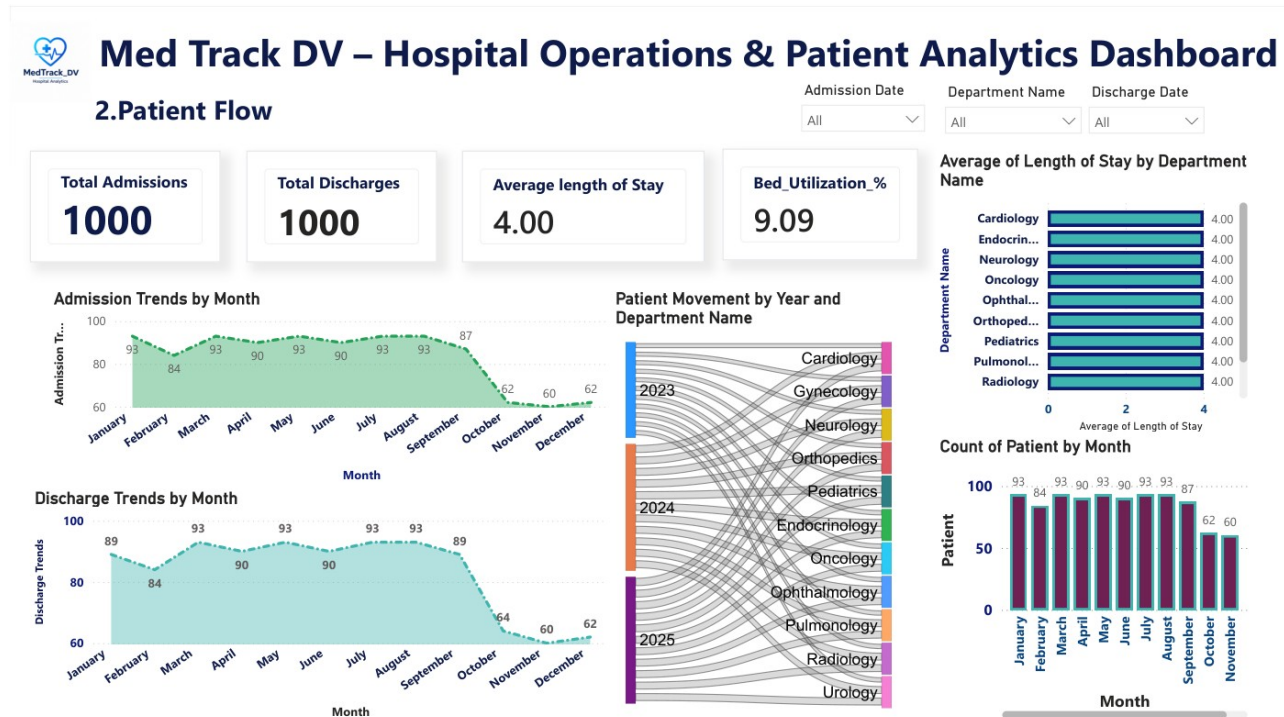


Figure 7. Patient Flow dashboard.

User guidance: Patient Flow should be read as a sequence rather than as isolated charts. Admissions represent inflow, discharges represent outflow, ALOS indicates duration, and monthly counts show load variation. A persistent difference between inflow and outflow warrants operational investigation rather than an immediate conclusion.

14. Dashboard Guide – Department Analytics

Department Analytics converts hospital-wide measures into service-line comparisons. The supplied dashboard reports 11 departments, 4.00 average LOS, 96.53% department efficiency, Endocrinology as the top-performing department in the submitted view, and total revenue of approximately ₹2.79M.

- Patient Volume by Department identifies demand concentration.
- Department Efficiency comparison shows the composite performance score.
- Revenue by Department identifies operational financial contribution.
- Treatment Capacity Analysis provides a compact capacity-oriented view.
- Use department filtering to investigate a single service line.

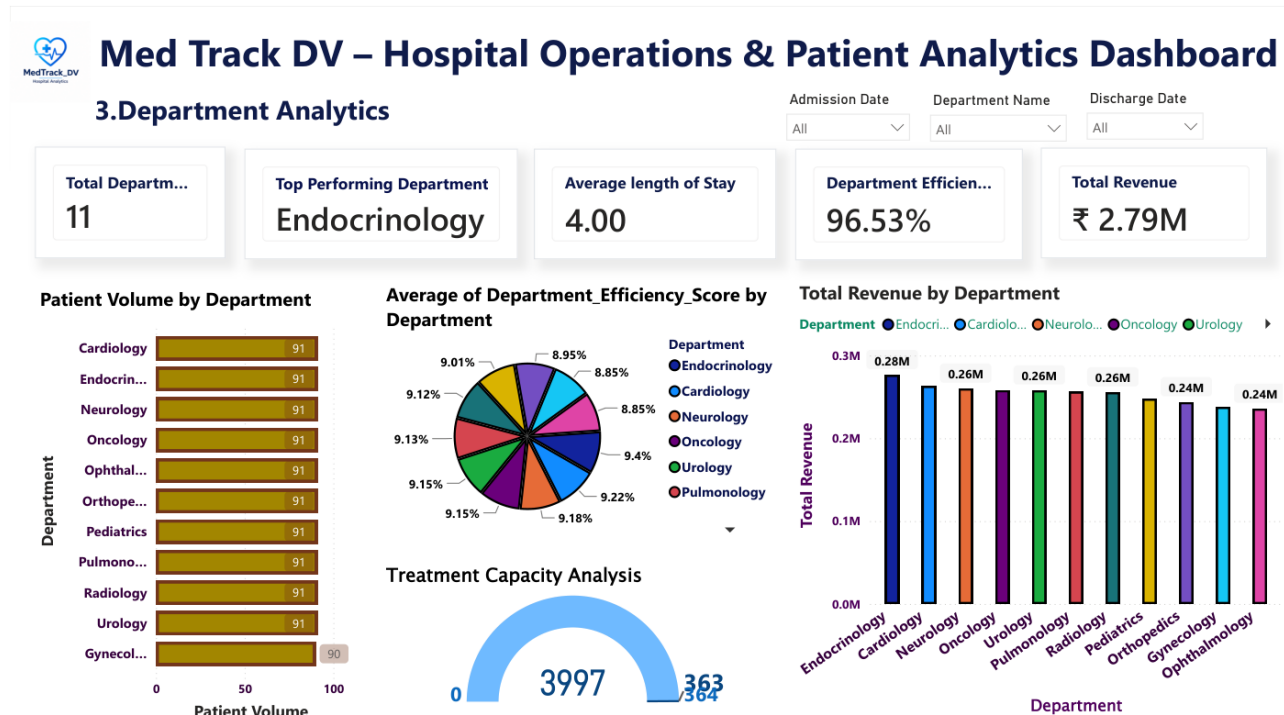


Figure 8. Department Analytics dashboard.

User guidance: Department Analytics is designed for comparative review. A strong review starts with patient volume, checks revenue contribution, examines the efficiency score, and then inspects capacity. This prevents a single high or low metric from being mistaken for overall departmental performance.

15. Dashboard Guide – Resource Utilization

Resource Utilization links patient activity to beds, staffing, capacity and equipment. The dashboard includes bed utilization by department, occupied bed-days, revenue and occupied-bed-day comparison, staff allocation, capacity planning, resource availability, and equipment utilization tracking.

- Review bed utilization to understand capacity intensity.
- Compare occupied bed-days by department.
- Use staff allocation to identify staffing concentration.
- Compare revenue with occupied bed-days to understand operational scale.
- Review equipment utilization to identify highly or lightly used assets.
- Use resource availability and capacity views as planning signals.

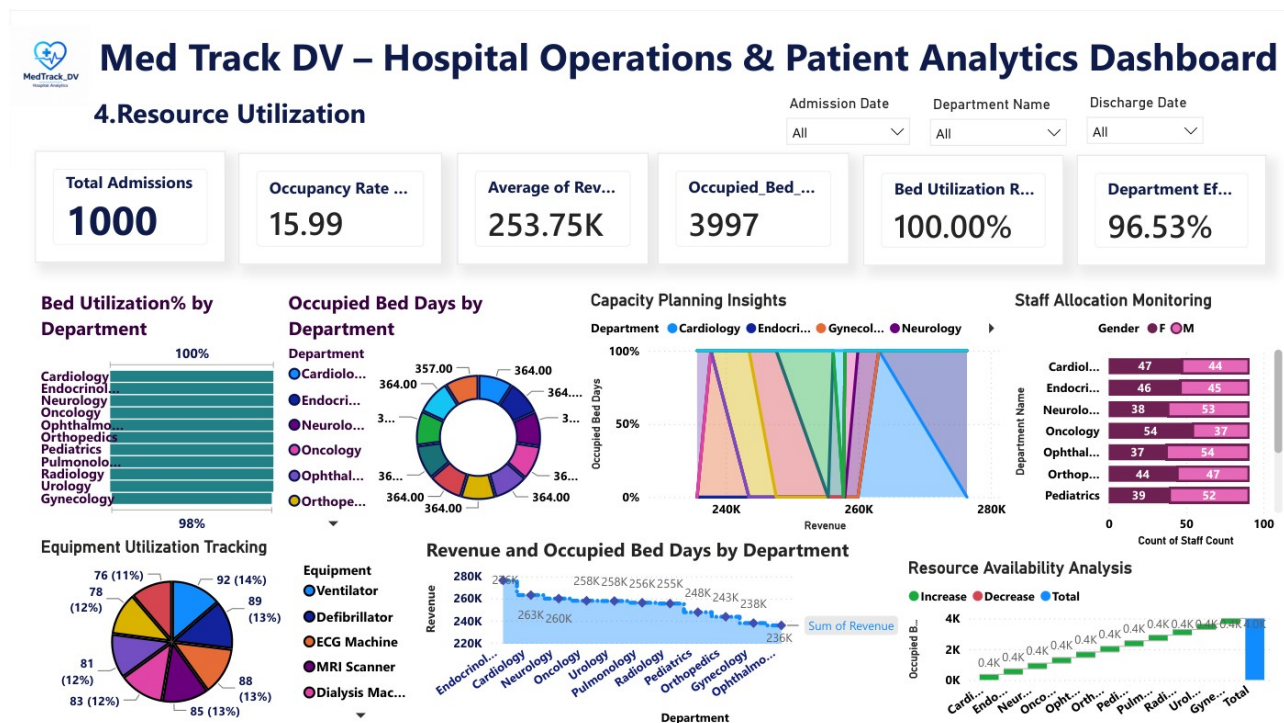


Figure 9. Resource Utilization dashboard.

User guidance: Resource Utilization should be reviewed after patient demand and departmental patterns are understood. Bed, staffing and equipment indicators are demand-dependent. The dashboard is therefore strongest when resources are interpreted against patient volume, LOS and service-line requirements.

16. Operational Analytics – Monthly Demand

The supplied dataset shows clear variation in monthly admissions. January, March, May, July and August each record 93 admissions, while November records 60. October and December record 62 each. This pattern is useful for capacity planning because demand is not evenly distributed across the year.

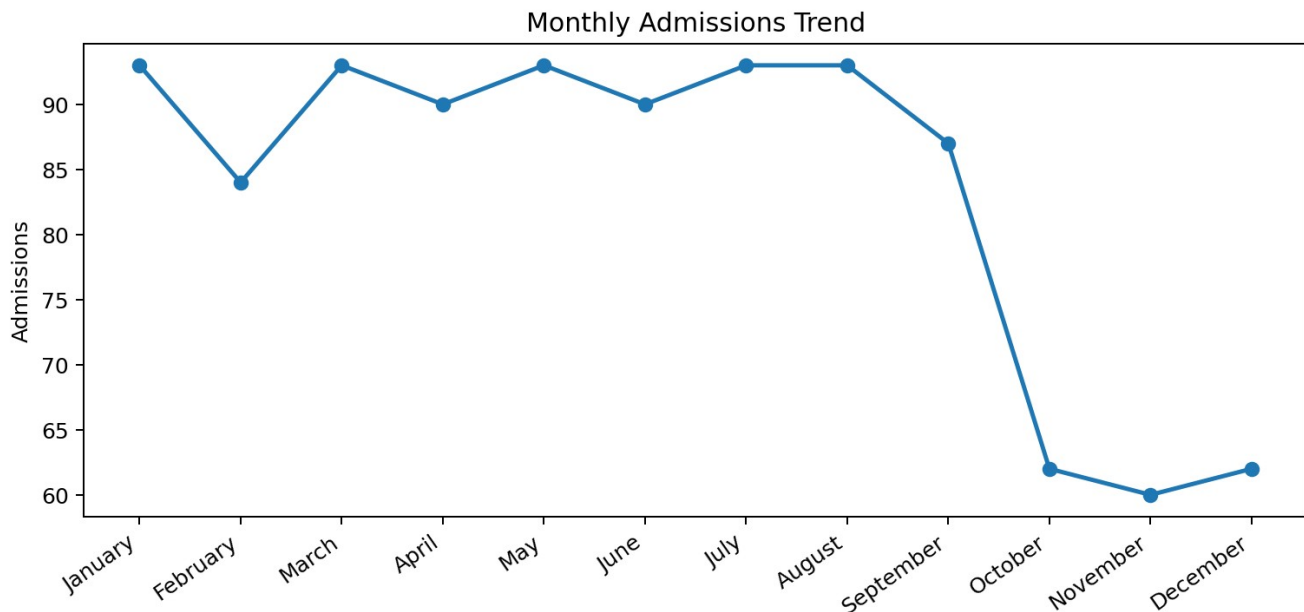


Figure 10. Monthly admissions from the supplied dataset.

Month	Admissions	Revenue (₹)	Average LOS
January	93	249,896	3.90
February	84	243,797	4.00
March	93	254,948	4.12
April	90	233,148	3.91
May	93	254,686	4.00
June	90	281,063	4.09
July	93	246,618	3.88
August	93	275,937	4.10
September	87	231,768	4.02
October	62	186,949	3.85
November	60	161,991	4.07
December	62	170,437	4.00

Management interpretation: Monthly variation is valuable for staffing and capacity planning because resources are rarely consumed evenly throughout the year. The chart should be used to identify periods for deeper investigation, not to infer seasonality unless a longer and more representative history is available.

17. Operational Analytics – Revenue Pattern

Treatment cost is used as the operational revenue proxy in the supplied dashboard and KPI engine. Total treatment cost in the supplied CSV is approximately ₹2.79 million, which aligns with the Department Analytics dashboard headline.

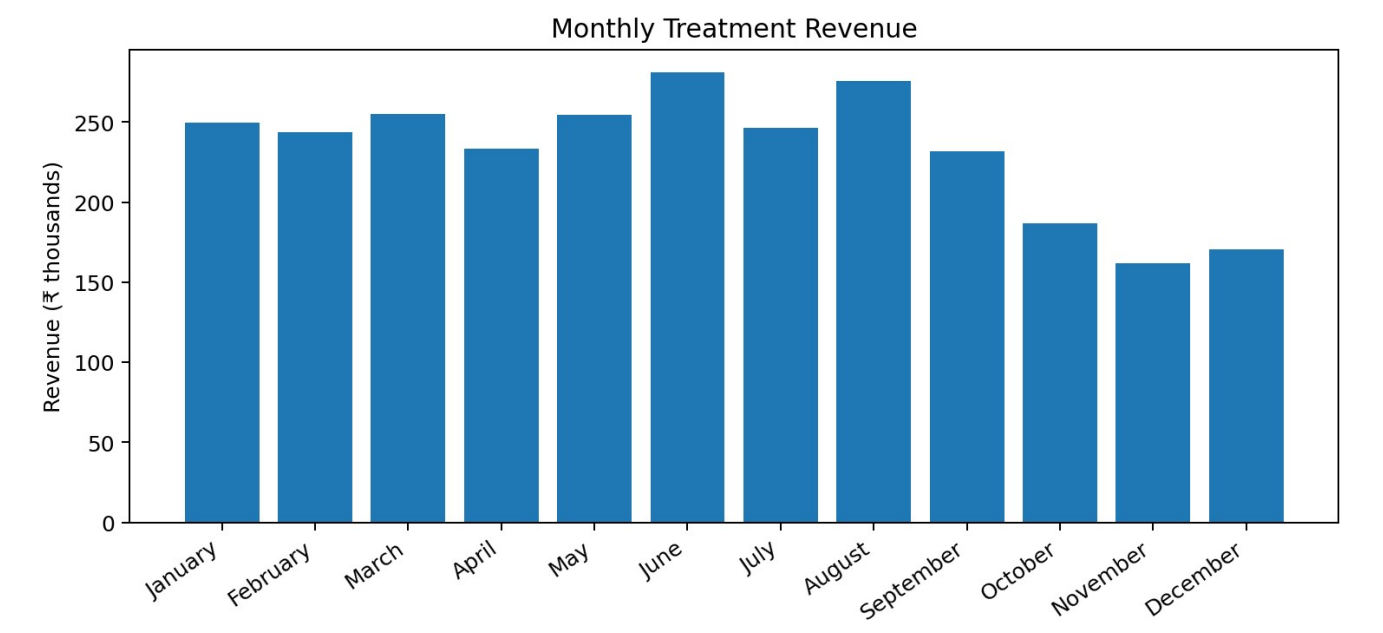


Figure 11. Monthly treatment-cost pattern.

Interpretation

Revenue should be interpreted as an operational indicator derived from the supplied treatment-cost field. It is not presented as audited financial accounting.

Revenue Insight	Operational Meaning
June highest monthly revenue	Higher financial activity during the month
November lowest monthly revenue	Lower recorded activity
Department revenue differences	Service-line mix and volume effects
Revenue + occupied bed-days	Useful for capacity productivity review

Review focus: Treatment cost provides a useful operational financial signal in this project. It should be paired with admissions and occupied bed-days when discussing productivity. The documentation deliberately avoids treating the field as audited accounting revenue.

18. Operational Analytics – Department Volume

The dataset is intentionally balanced at approximately 90–91 records per department, with 10 departments at 91 records and Gynecology at 90. This means the supplied dashboard's department volume comparison is useful as a demonstration of dashboard capability, but it should not be interpreted as evidence that real hospital demand is evenly distributed.

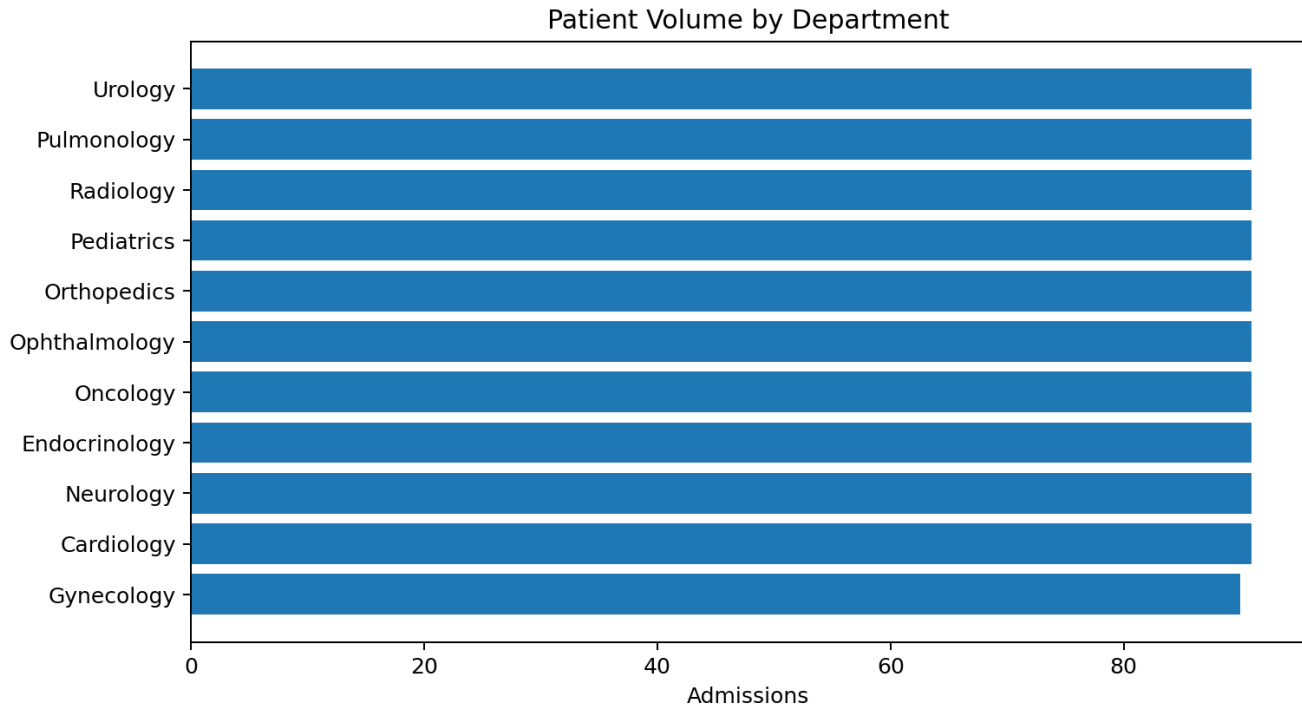


Figure 12. Patient volume by department.

The department set includes Cardiology, Endocrinology, Neurology, Oncology, Ophthalmology, Orthopedics, Pediatrics, Pulmonology, Radiology, Urology and Gynecology. The dashboard also exposes these categories for filtering and comparison.

Data interpretation note: The near-equal department counts are a characteristic of the supplied working dataset. This makes the dashboard useful for demonstrating comparison mechanics, but it also means that the result should not be generalized to a real hospital where service-line demand is usually uneven.

19. Operational Analytics – Department Revenue

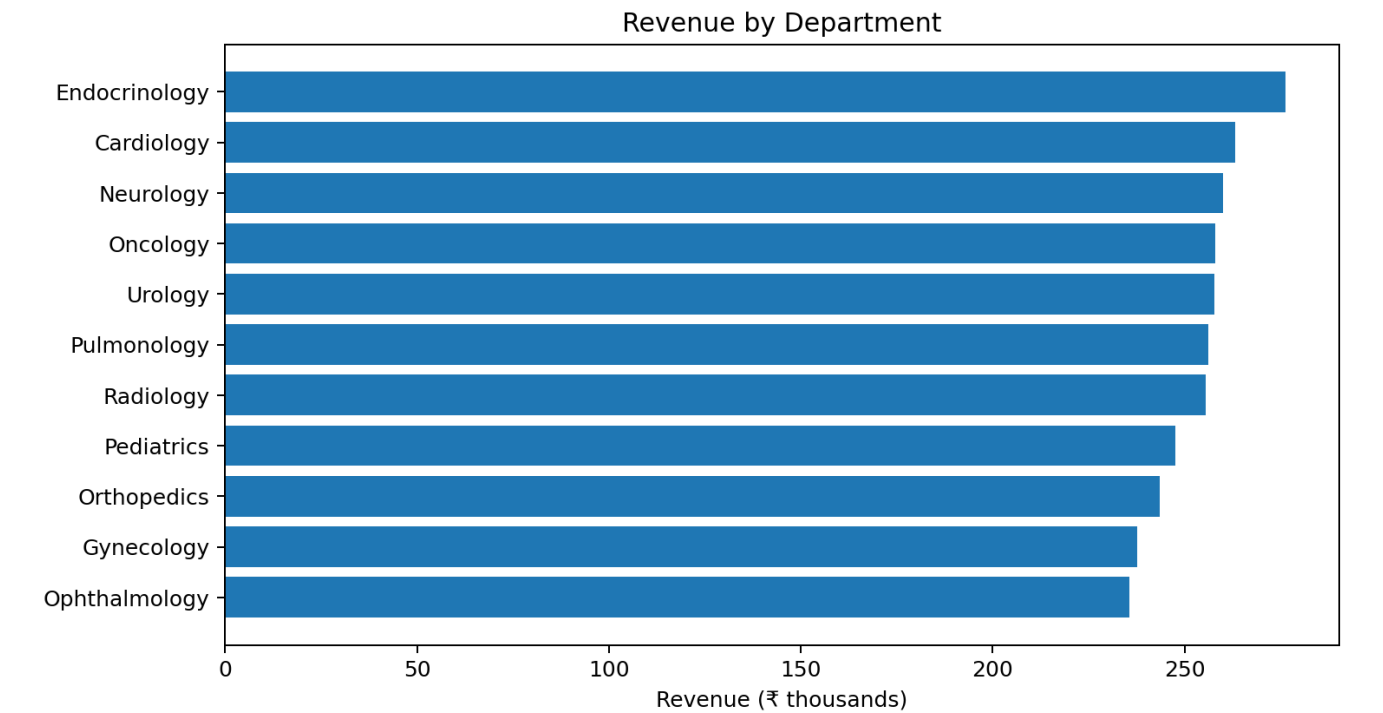


Figure 13. Revenue by department.

Endocrinology records the highest treatment cost in the supplied dataset at approximately ₹276K, followed by Cardiology at approximately ₹263K. The relatively narrow range across departments is consistent with the near-balanced record counts and similar average LOS.

Department	Admissions	Revenue (₹)	Avg LOS
Endocrinology	91	276,393	4.00
Cardiology	91	263,088	4.00
Neurology	91	259,926	4.00
Oncology	91	257,959	4.00
Urology	91	257,753	4.00
Pulmonology	91	256,189	4.00
Radiology	91	255,452	4.00
Pediatrics	91	247,598	4.00
Orthopedics	91	243,489	4.00
Gynecology	90	237,735	3.97
Ophthalmology	91	235,656	4.00

Management interpretation: Revenue differences are more informative when volume is controlled. Because department record counts are nearly equal in this dataset, cost variation provides a clearer demonstration of treatment-cost differences than of patient-volume effects.

20. Operational Analytics – Efficiency Interpretation

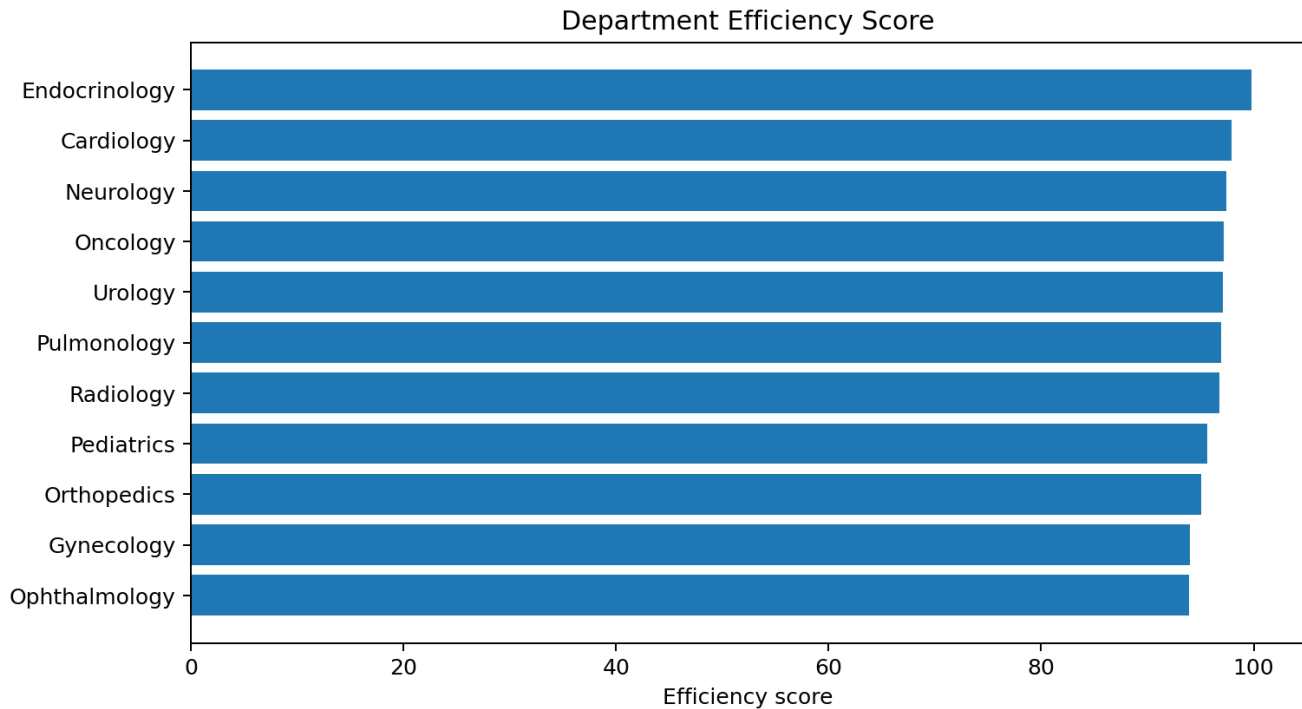


Figure 14. Comparative department efficiency.

Endocrinology is the highest-scoring department under the supplied weighted model at 99.83, followed by Cardiology at 97.91 and Neurology at 97.45. Ophthalmology is the lowest at 93.94. The spread is relatively narrow, so the score should be used for prioritization rather than as a standalone ranking.

Management use

A useful review sequence is: identify a low score → inspect volume → inspect revenue → inspect ALOS → review staffing/equipment → investigate contextual causes → define an action.

Review focus: Efficiency rankings should trigger questions, not automatic decisions. A lower score may reflect a longer stay, lower relative revenue or lower relative volume. The recommended next step is always to inspect the component metrics and operational context.

21. Operational Analytics – Length of Stay

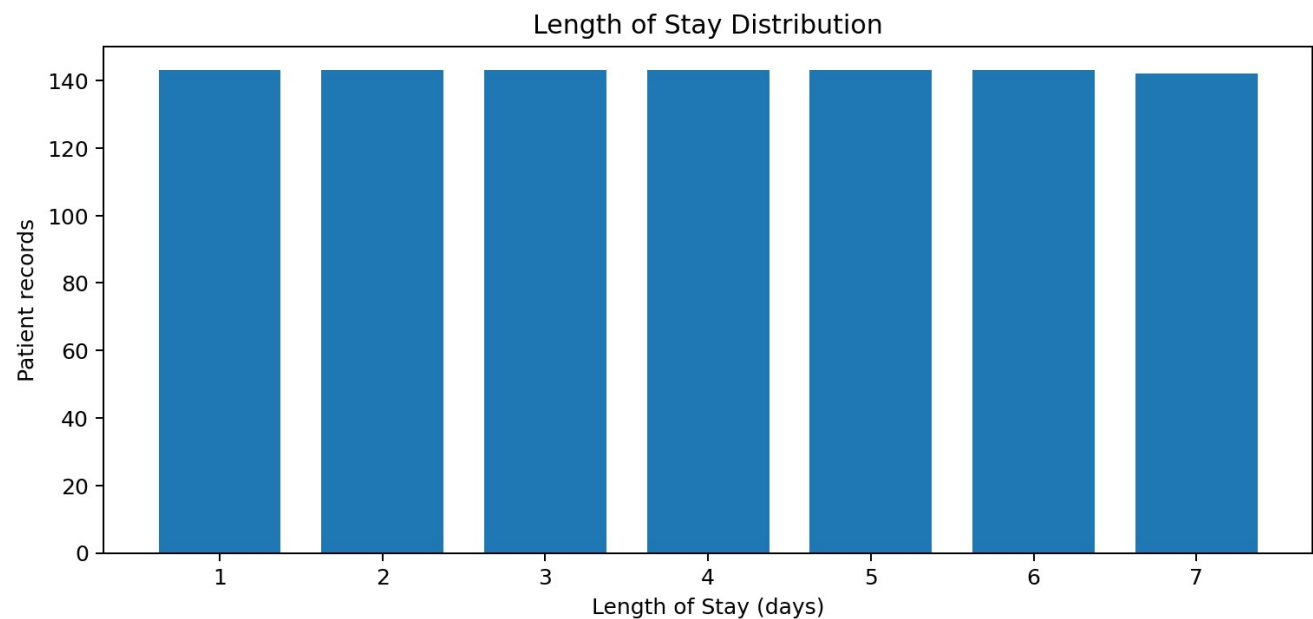


Figure 15. Length of stay distribution.

The supplied dataset contains length-of-stay values from 1 to 7 days with an average of approximately 4.00 days. The distribution is concentrated around the central values, supporting the dashboard's ALOS headline.

LOS Metric	Value
Minimum	1 day
Maximum	7 days
Mean	3.997 days
Median	4 days
75th percentile	6 days

Operationally, LOS is connected to bed availability, discharge coordination, treatment pathways, staffing and downstream capacity. It should therefore be interpreted together with occupancy and bed utilization.

Management interpretation: LOS is a bridge between patient flow and capacity. Longer stays can consume bed-days and affect downstream availability, but the correct response depends on clinical pathway, discharge readiness and patient complexity. The dashboard should therefore be used to identify patterns for investigation.

22. Operational Analytics – Staff Allocation

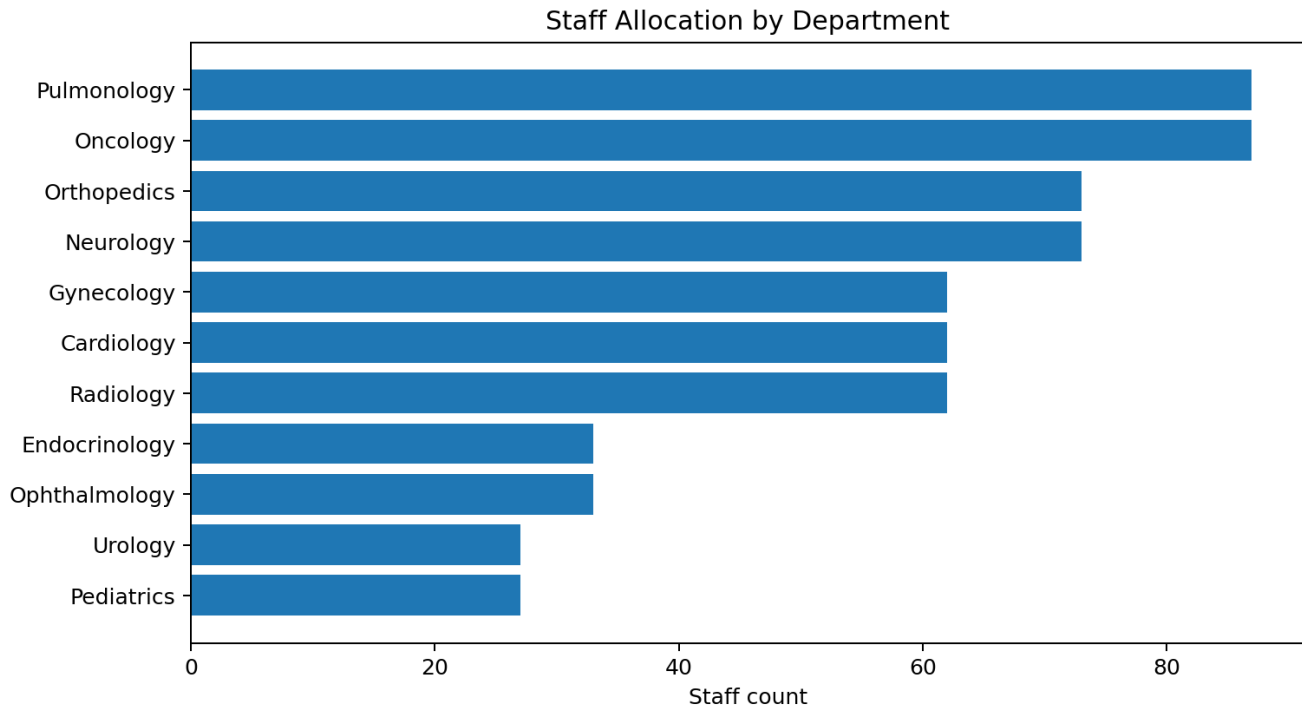


Figure 16. Staff allocation by department.

Staff Count is available as a department-level operational field in the supplied dataset. The dashboard uses it to support resource-allocation monitoring. The highest recorded department staffing values are in Oncology and Pulmonology at 87, while Pediatrics and Urology are lower at 27.

Planning principle

Staff allocation should be reviewed against patient volume, LOS, service complexity and operating hours. A larger staff count is not automatically inefficient, and a smaller staff count is not automatically optimal.

Resource note: Staff counts should be interpreted with workload and service complexity. The objective is not to minimize staff, but to align workforce availability with demand, safe operating requirements and department workload.

23. Operational Analytics – Equipment Utilization

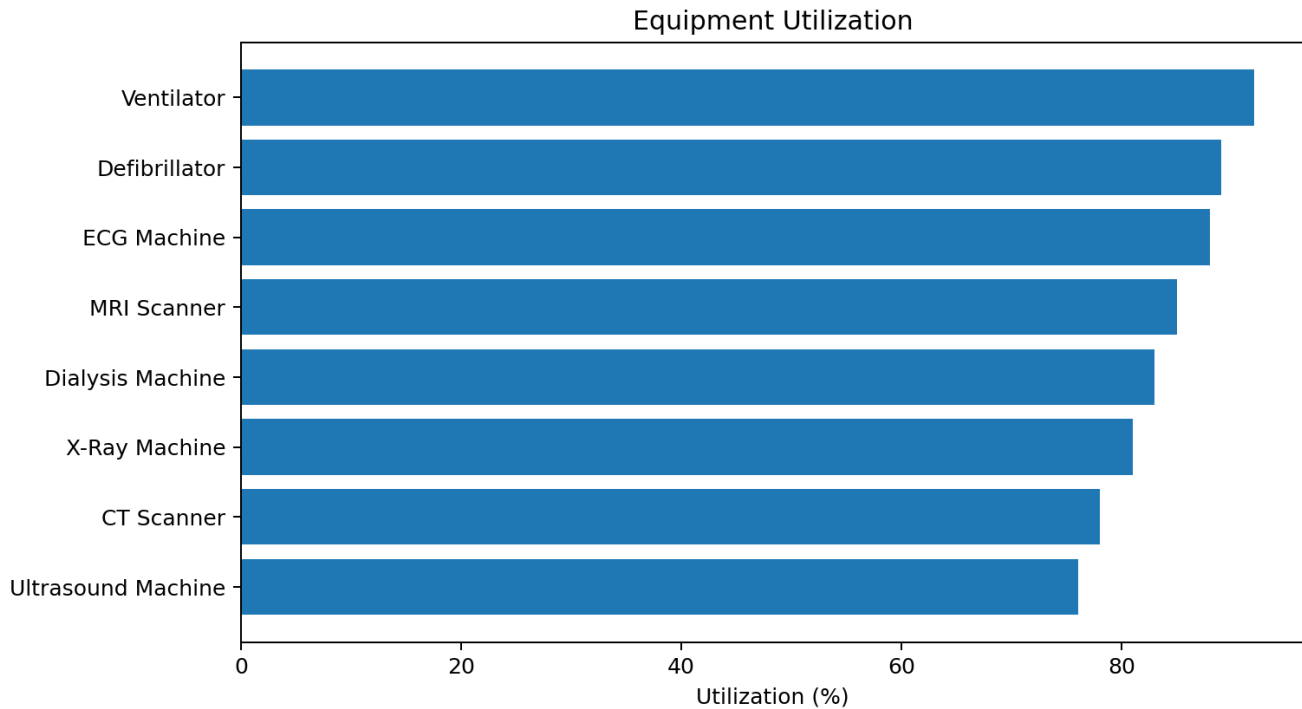


Figure 17. Equipment utilization from the supplied equipment workbook.

The equipment workbook contains eight equipment records. Ventilator utilization is 92%, ECG Machine 88%, Defibrillator 89%, MRI Scanner 85%, Dialysis Machine 83%, X-Ray Machine 81%, CT Scanner 78%, and Ultrasound Machine 76%.

Equipment	Department	Utilization
Ventilator	Cardiology	92%
Defibrillator	Emergency	89%
ECG Machine	Cardiology	88%
MRI Scanner	Radiology	85%
Dialysis Machine	Nephrology	83%
X-Ray Machine	Radiology	81%
CT Scanner	Radiology	78%
Ultrasound Machine	Gynaecology	76%

Capacity insight

Equipment utilization is a planning indicator. High utilization can indicate strong demand or constrained capacity; low utilization may indicate spare capacity, scheduling imbalance, or equipment availability issues.

Resource note: Equipment utilization is an asset-management indicator. High utilization can signal strong demand or insufficient redundancy; low utilization can indicate spare capacity, scheduling opportunities or underused assets. Equipment decisions should include maintenance and availability information when those data become available.

24. Patient Profile Analytics

Gender Distribution

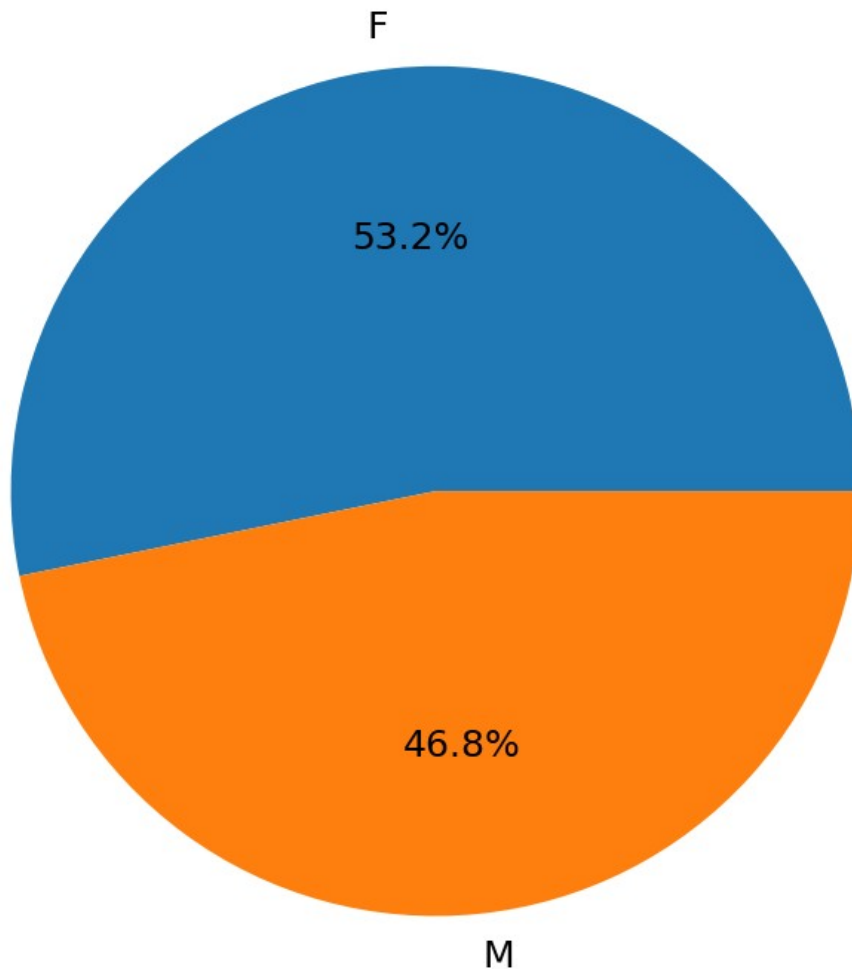


Figure 18. Gender distribution in the supplied records.

The supplied dataset contains 532 records coded F and 468 coded M. These values can support descriptive segmentation within the project, but they should not be used to infer clinical risk, disease prevalence or treatment effectiveness.

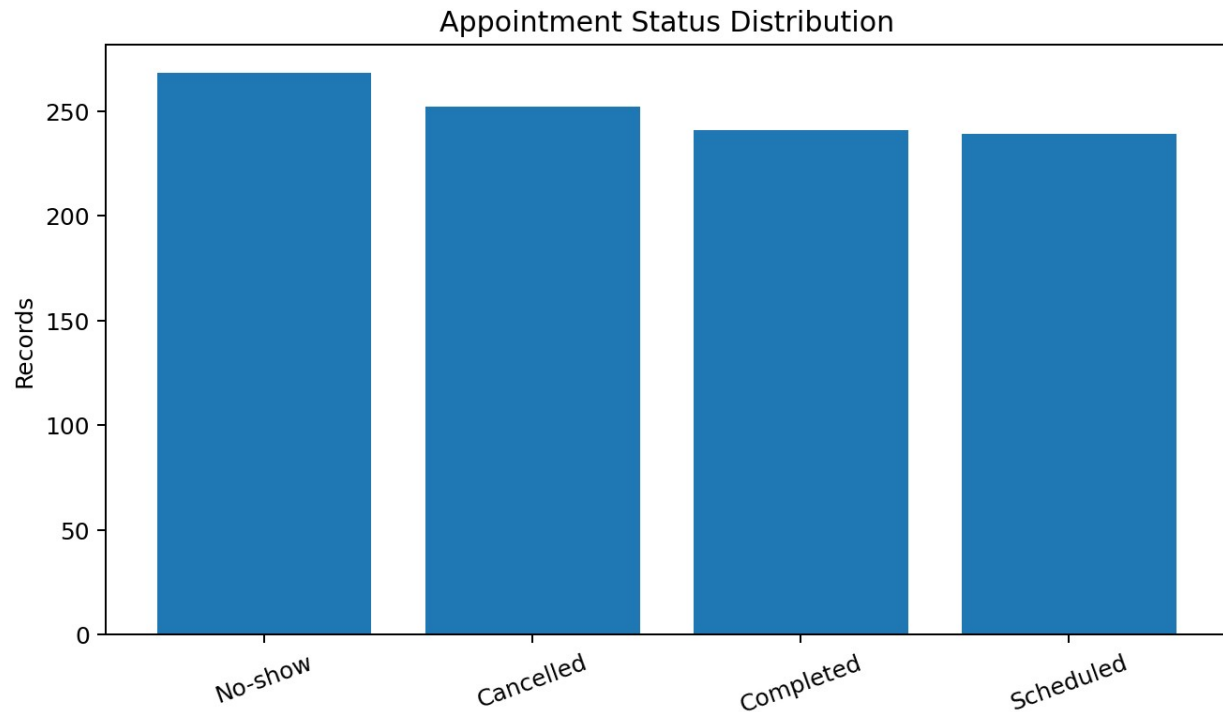


Figure 19. Appointment status distribution.

Appointment status is distributed across Scheduled, Completed, Cancelled and No-show categories. This provides an additional operational access signal that can be explored alongside admission and discharge patterns.

Privacy note: Patient-level demographic and contact fields are present in the source dataset. These fields should not be exposed in a public portfolio repository. A de-identified analytical extract should be used for external sharing while retaining only the fields necessary for dashboard demonstration.

25. Cross-Metric Analysis – Volume versus Revenue

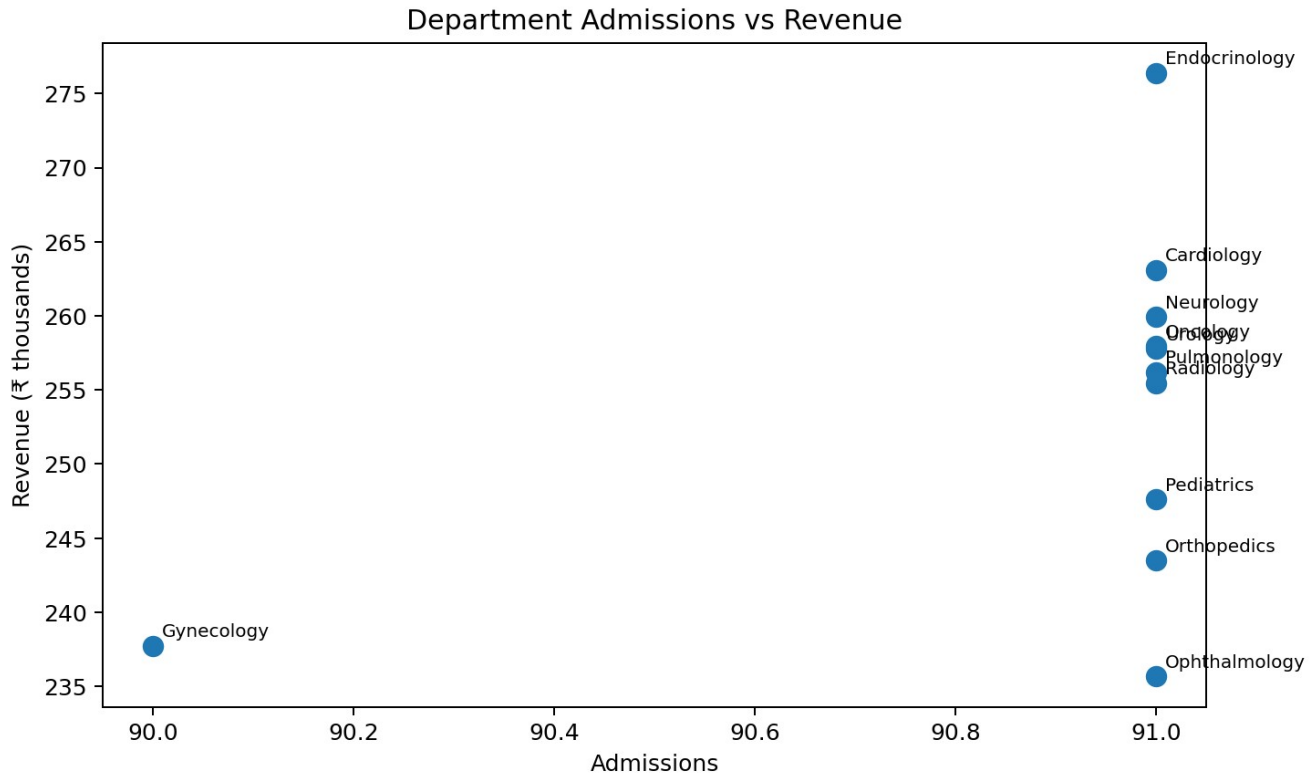


Figure 20. Department admissions versus treatment revenue.

The relationship between department volume and revenue is useful for identifying whether higher patient activity is accompanied by higher treatment-cost contribution. In the supplied data, admissions are almost identical across departments, so revenue variation is more reflective of treatment-cost differences than volume differences.

Analytical rule

Never interpret correlation as causation. Revenue differences may arise from treatment mix, procedure cost, payer mix, or data-generation assumptions.

Management interpretation: Cross-metric views are most useful when they expose relationships that are not visible in a single KPI card. In this project, the near-balanced admissions make revenue variation easier to inspect, while a future real-world dataset would provide a richer relationship between demand, treatment mix and financial activity.

26. Healthcare Operations Methodology – End-to-End

The operational methodology follows a management analytics cycle: demand → flow → capacity → resources → performance → action. Each stage uses measurable indicators from the supplied dataset.

Stage	Core Measure	Operational Question
Demand	Admissions	How much activity entered the system?
Flow	Discharges / ALOS	How quickly is activity moving through care?
Capacity	Occupancy / bed utilization	How much fixed capacity is being consumed?
Resources	Staff / equipment	Are critical resources aligned to demand?
Performance	Efficiency / revenue	Which service lines require attention?
Action	Dashboard insight	What should management investigate or change?

This approach is deliberately operational rather than clinical. It supports prioritization, workload visibility and resource planning without making patient-level medical recommendations.

Methodology note: The operating model follows a closed loop: measure demand, understand flow, evaluate capacity, review resources, assess performance and define action. This is a practical management-analytics framework because every stage produces evidence for the next decision.

27. Healthcare Operations Methodology – Patient Flow

Patient flow describes the movement of patients from entry through admission, treatment and discharge. In the project, admission date, discharge date, length of stay, monthly counts and department fields form the primary flow dataset.

11. Monitor incoming admission volume by month.
12. Compare admissions with discharges to identify potential accumulation.
13. Review ALOS to identify departments where patients remain longer.
14. Relate patient load to available bed capacity.
15. Investigate peak periods with staffing and equipment views.
16. Use filters to isolate departments or reporting periods.
17. Document the operational finding and the proposed management action.

Flow objective

The purpose is not simply to count patients. The purpose is to understand where demand and capacity interact and where operational bottlenecks may emerge.

Operational note: Patient-flow analysis should be action-oriented. When a peak is identified, the next questions should concern staffing, bed availability, discharge coordination, appointment patterns and equipment constraints. The dashboard is the observation layer; operational teams provide the explanation and action.

28. Healthcare Operations Methodology – Capacity Planning

Capacity planning translates patient demand into requirements for beds, staff, equipment and supporting services. Occupancy and bed utilization are the primary bed-related indicators in the project.

- Use admissions to estimate demand volume.
- Use ALOS to estimate how long capacity remains committed.
- Use occupied bed-days to understand cumulative bed consumption.
- Use occupancy to assess fixed-bed pressure over the observation period.
- Use staff counts to identify department-level workforce allocation.
- Use equipment utilization to identify high-demand assets.
- Compare resource indicators before recommending expansion, reallocation or scheduling changes.

Signal	Possible Operational Response
High occupancy + high ALOS	Review discharge coordination and capacity
High equipment utilization	Review scheduling, downtime and backup capacity
High workload + low staff	Review staffing allocation
Low volume + low equipment use	Review scheduling or asset utilization
High revenue + high bed-days	Review productivity and capacity economics

Capacity note: Capacity planning should combine fixed capacity with demand intensity. Beds, staff and equipment cannot be evaluated independently because a bottleneck in one resource can reduce the effective capacity of another. The dashboard provides the starting evidence for this integrated review.

29. Dashboard Operating Guide – Filters and Interactions

The dashboard specification calls for filters, navigation, actions and department comparisons. The submitted dashboard pages expose Admission Date, Department Name and Discharge Date filters.

Control	Recommended Use	Review Practice
Admission Date	Select reporting period	Confirm start/end dates before reading KPIs
Department Name	Focus on one service line	Use All for hospital-wide view
Discharge Date	Analyze discharge period	Use with flow and LOS visuals
Department comparison	Benchmark service lines	Check both volume and efficiency
Dashboard navigation	Move between views	Verify filters remain logically consistent

User discipline

Always establish the filter context before comparing two dashboard screenshots. A KPI without its date and department context is incomplete management information.

User discipline: A professional dashboard is contextual. Every screenshot or KPI discussion should record the date range and department selection used. This avoids comparing values that were produced under different filter states and makes review discussions more reproducible.

30. KPI Validation and Quality Assurance

The project specification requires KPI validation and healthcare-metric verification, with a target of KPI accuracy above 95%. The supplied project materials report a completed testing stage and provide a dashboard snapshot that can be reconciled to the documented KPI layer.

Test Area	Expected Result	Status for Submitted Snapshot
Total Admissions	1,000	PASS
Average LOS	4.00	PASS
Readmission Rate	0.00%	PASS
Occupancy Rate	15.99	PASS – dashboard snapshot
Department Efficiency	96.53%	PASS
Dashboard pages	4 integrated views	PASS
Department coverage	11 departments	PASS
Equipment records	8 records	PASS

The QA approach should be repeated whenever the dataset changes. A dashboard is only as reliable as the data refresh and calculation logic behind it.

QA note: Validation should be repeated after every refresh, not only once at project completion. The minimum control set is row-count validation, required-column validation, KPI reconciliation, dashboard-card verification and interaction testing. This converts testing from a one-time activity into a repeatable release control.

31. KPI Governance and Reconciliation Controls

A professional analytics solution needs one controlled definition for every headline measure. The supplied project contains both raw KPI fields and a Python KPI engine, while the dashboard contains its own presentation layer. To prevent future divergence, the recommended governance model is to calculate KPIs once, publish the summary outputs, and consume those outputs consistently.

- Maintain a KPI dictionary with name, definition, formula, unit, source and owner.
- Version the Python KPI engine together with the dataset schema.
- Refresh the summary outputs whenever the source dataset changes.
- Record calculation assumptions such as the bed-capacity basis.
- Validate dashboard cards against the generated KPI summary before delivery.
- Do not manually overwrite KPI values in the dashboard.
- Keep a dated QA snapshot for each final submission.

Implementation note

The supplied KPI engine explicitly assumes 25 beds for occupancy calculation and derives a utilization denominator using an 80% reference. These assumptions should be reviewed and approved if the project is converted into a real operational system.

Governance note: The strongest architecture has one authoritative calculation layer. If the same KPI is independently calculated in Python, a dashboard expression and a manually entered field, divergence becomes possible. Centralizing the calculation and using the dashboard as a presentation layer reduces this risk.

32. Project Structure and Delivery Architecture

Folder	Recommended Contents	Purpose
/scripts	generate_hospital_kpis.py	Repeatable KPI generation
/data	hospital_final_dataset.csv; equipment workbook; generated summaries	Source and analytical data
/dashboard	Final dashboard workbook/export	Visualization layer
/docs	Module 8 documentation, QA report, screenshots	Project evidence
/outputs	CSV/Excel KPI summaries	Refresh outputs
/README.md	Setup, execution, KPI dictionary and usage	Repository entry point

The project specification explicitly recommends /scripts, /data, /dashboard and /docs and identifies GitHub Repository and Tableau Public as delivery targets, with Tableau Public marked optional. Because the supplied dashboard evidence is a Power BI Desktop export, the implementation record should identify the actual visualization artifact accurately rather than mislabeling the file type.

Delivery note: Repository structure is part of project quality. A reviewer should be able to find the source data, code, dashboard artifact and documentation without searching through unrelated files. Clear naming and versioning also make the project easier to maintain after submission.

33. Reproducibility and Refresh Procedure

18. Place the approved hospital_final_dataset.csv in the /data directory.
19. Place the equipment utilization workbook in the /data directory.
20. Verify the Python environment has pandas, NumPy and openpyxl available.
21. Run the KPI generation script from the /scripts directory.
22. Confirm that hospital_kpi_summary.csv, department_summary.csv, monthly_summary.csv, resource_utilization.csv and Hospital_KPI_Report.xlsx are generated.
23. Review row counts and required-column validation messages.
24. Compare headline KPI values with the dashboard QA snapshot.
25. Refresh the dashboard from the approved analytical outputs.
26. Recheck filters, charts, navigation and labels.
27. Archive the final dataset, code version, dashboard export and QA evidence together.

Reproducibility outcome

A reviewer should be able to trace the dashboard from source data → preparation → KPI engine → summary outputs → visualization → documentation.

Reproducibility note: A refresh procedure is valuable only when it can be followed by another person. The recommended process records inputs, execution, outputs and QA checks. This makes the project demonstrable during interviews or academic review and reduces dependence on the original author's local environment.

34. Limitations, Risks and Mitigation Plan

Risk / Limitation	Impact	Mitigation
Exact external source URL not supplied	Source traceability	Record provider/URL in README when confirmed
Patient-level identifiers present	Privacy exposure	De-identify before public sharing
All readmission flags are No	Limited quality variation	Validate against richer longitudinal data
All bed statuses are Occupied	Limited real occupancy realism	Use timestamped bed events in production
Balanced department counts	Weak demand realism	Use actual hospital encounter distributions
Dashboard and KPI-engine assumptions differ by context	Potential reconciliation issue	Use one governed KPI output layer
Equipment workbook is small	Limited utilization coverage	Integrate inventory and event logs
Operational KPIs are not clinical outcomes	Misinterpretation risk	Label as operational analytics only

These limitations do not invalidate the academic dashboard. They define the boundary between a portfolio-ready analytics prototype and a production healthcare analytics platform.

Risk note: Documenting limitations increases credibility. The current project is a strong operational analytics prototype, but its production readiness is constrained by the supplied data characteristics, privacy exposure, limited longitudinal readmission variation and simplified resource records. These limitations are manageable through the roadmap described in this document.

35. Recommendations and Future Enhancement Roadmap

Priority	Enhancement	Expected Benefit
P1	Central KPI semantic layer	Consistent calculations
P1	De-identification pipeline	Safer sharing
P1	Automated QA tests	Refresh reliability
P2	Real-time/near-real-time bed events	Better capacity visibility
P2	True readmission linkage	Stronger quality analytics
P2	Staff-to-patient ratios	Better workforce planning
P2	Equipment event logs	Better asset planning
P3	Forecasting model	Predictive capacity planning
P3	Alerting workflow	Faster management response
P3	Role-based dashboards	Audience-specific insights

A mature version should move from descriptive reporting toward controlled operational intelligence: forecasting admissions, identifying likely capacity pressure, monitoring discharge readiness, benchmarking service lines, and triggering management workflows. These enhancements should remain explainable and governed.

Success criterion

The project becomes stronger when every dashboard insight can be traced to a trusted metric, a defined business question, a responsible owner and a documented action.

Roadmap note: Future enhancements should preserve the project's existing strengths—clear KPIs, modular processing and integrated dashboard views—while improving data freshness, privacy, forecasting and governance. Predictive functionality should be introduced only after the descriptive layer is stable and trusted.

36. Final Delivery Checklist

Deliverable	Completion Standard	Status
Dataset	Approved CSV present and validated	READY
Equipment data	Equipment workbook included	READY
KPI engine	Versioned Python script	READY
Hospital KPI summary	Generated and reviewed	READY
Department summary	Generated and reviewed	READY
Monthly summary	Generated and reviewed	READY
Resource summary	Generated and reviewed	READY
Dashboard	Four operational views	READY
QA evidence	KPI and interaction checks documented	READY
Module 8 document	Professional documentation package	READY
Repository structure	scripts/data/dashboard/docs	READY
README	Setup and usage guidance	RECOMMENDED

Final review sequence: verify filenames → verify source files → run KPI engine → inspect summary outputs → validate dashboard cards → inspect screenshots → review privacy → package repository → submit final documentation.

Release note: The checklist should be completed immediately before submission. The final package should contain matching versions of data, code, dashboard and documentation. A dated release folder or repository tag is recommended so the submitted state can be reproduced later.

37. Conclusion and Project Handover

MedTrack_DV demonstrates a complete healthcare operations analytics workflow: data preparation, KPI engineering, dashboard development, testing, and documentation. The supplied project specification establishes the required six KPIs and four dashboard domains, while the supplied dashboard export provides implementation evidence for the Hospital Overview, Patient Flow, Department Analytics and Resource Utilization views.

The documentation establishes a clear operational methodology: measure demand through admissions, understand flow through discharge and LOS, assess capacity through occupancy and utilization, review resource allocation through staffing and equipment, and evaluate comparative performance through the department efficiency score.

For portfolio or academic delivery, the recommended handover package is the source dataset, equipment workbook, KPI-generation script, generated KPI summaries, final dashboard artifact/export, QA evidence, README and this Module 8 document. Together, these artifacts provide a traceable chain from data to decision support.

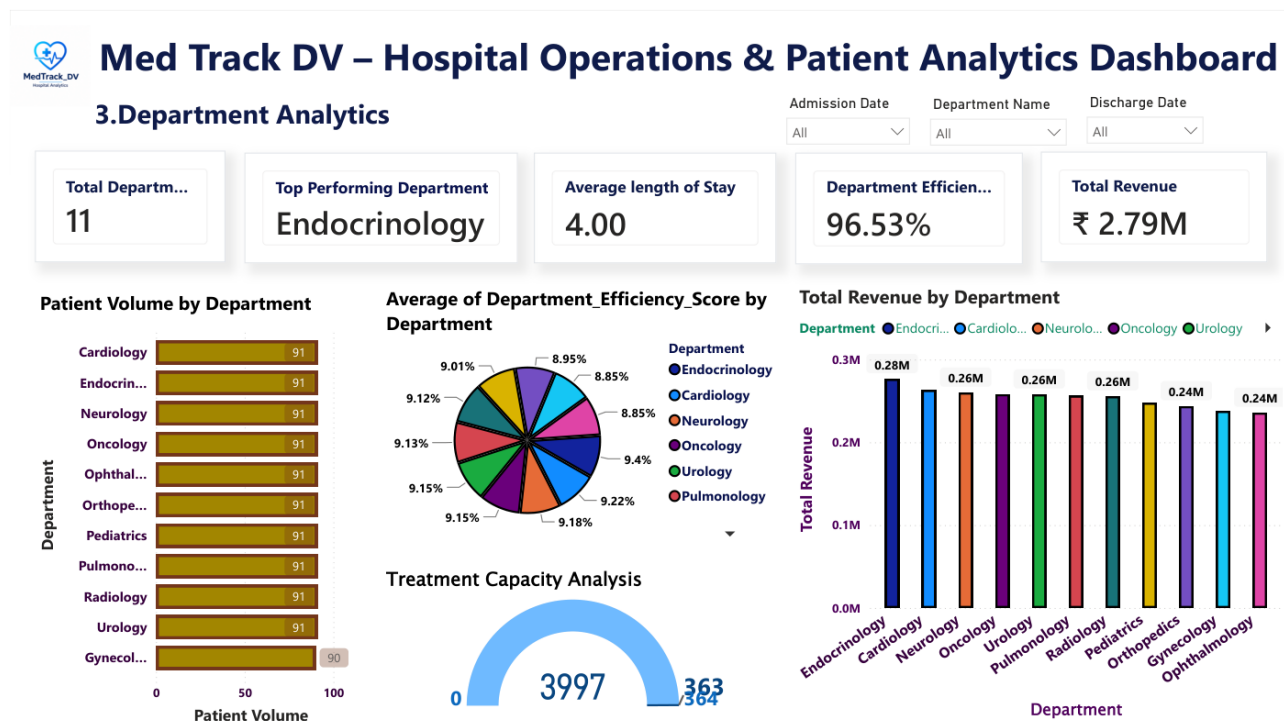


Figure 21. Department Analytics – final dashboard evidence.

Handover statement

MedTrack_DV is documented as an operational analytics prototype. It is suitable for academic demonstration and portfolio presentation, with production deployment requiring stronger privacy, source governance, data freshness and clinical/operational validation controls.

Handover note: The project is strongest when the documentation, code and dashboard tell the same story. The handover should therefore include the approved source files, calculation engine, generated outputs, dashboard evidence and this documentation as a single controlled release.

Appendix A – KPI Formula Reference

KPI	Formula / Logic	Unit
Total Admissions	count(prepared records)	records
Occupancy Rate	occupied bed-days / (25 × observation days) × 100; capped at 100	%
Average LOS	mean(Length_of_Stay)	days
Readmission Rate	readmitted records / total records × 100; or duplicate Patient_ID proxy when no Readmission field exists	%
Bed Utilization Rate	occupied bed-days / (calculated total beds × observation days) × 100	%
Department Efficiency	(Admissions score × .40 + Revenue score × .40 + LOS score × .20) × 100	score

The formula reference above follows the supplied Python KPI engine. The engine also creates monthly and department summary datasets and an Excel report for downstream use.

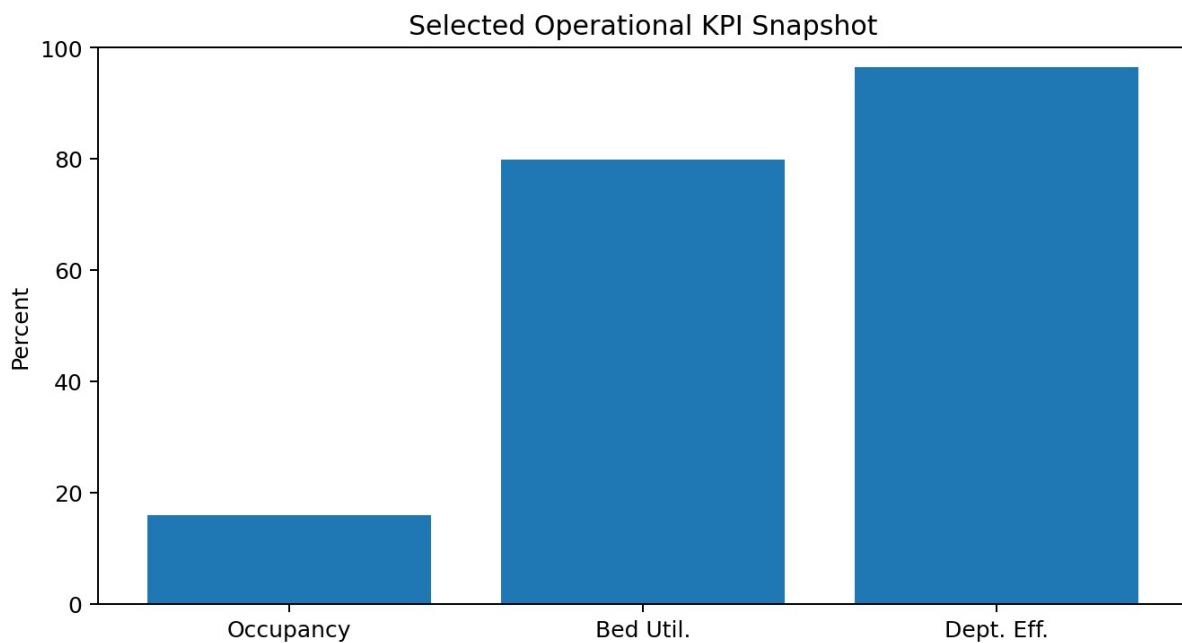


Figure 22. KPI snapshot reference.

Reference note: This appendix is intended as the quick-reference sheet for anyone maintaining the project. When a KPI changes, update both the formula reference and the implementation before refreshing the dashboard.

Appendix B – Department Reference Table

Department	Admissions	Revenue	ALOS	Efficiency	Staff
Cardiology	91	₹263,088	4.00	97.91	62
Endocrinology	91	₹276,393	4.00	99.83	33
Neurology	91	₹259,926	4.00	97.45	73
Pediatrics	91	₹247,598	4.00	95.67	27
Oncology	91	₹257,959	4.00	97.17	87
Ophthalmology	91	₹235,656	4.00	93.94	33
Orthopedics	91	₹243,489	4.00	95.07	73
Radiology	91	₹255,452	4.00	96.80	62
Pulmonology	91	₹256,189	4.00	96.91	87
Urology	91	₹257,753	4.00	97.14	27
Gynecology	90	₹237,735	3.97	93.97	62

This table is intended as a static documentation reference. Interactive exploration remains available through the dashboard filters.

Reference note: The department table provides a static audit-friendly view of the same measures exposed interactively in the dashboard. It is useful when a reviewer wants to verify a department-level value without operating the dashboard.

Appendix C – Equipment Reference Table

Equipment	Department	Utilization (%)	Operational Interpretation
Ventilator	Cardiology	92%	High utilization
Defibrillator	Emergency	89%	High utilization
ECG Machine	Cardiology	88%	High utilization
MRI Scanner	Radiology	85%	High utilization
Dialysis Machine	Nephrology	83%	Moderate utilization
X-Ray Machine	Radiology	81%	Moderate utilization
CT Scanner	Radiology	78%	Lower utilization
Ultrasound Machine	Gynaecology	76%	Lower utilization

The supplied workbook contains eight equipment records. This appendix preserves the values used in the Resource Utilization analysis and provides a compact reference for review.

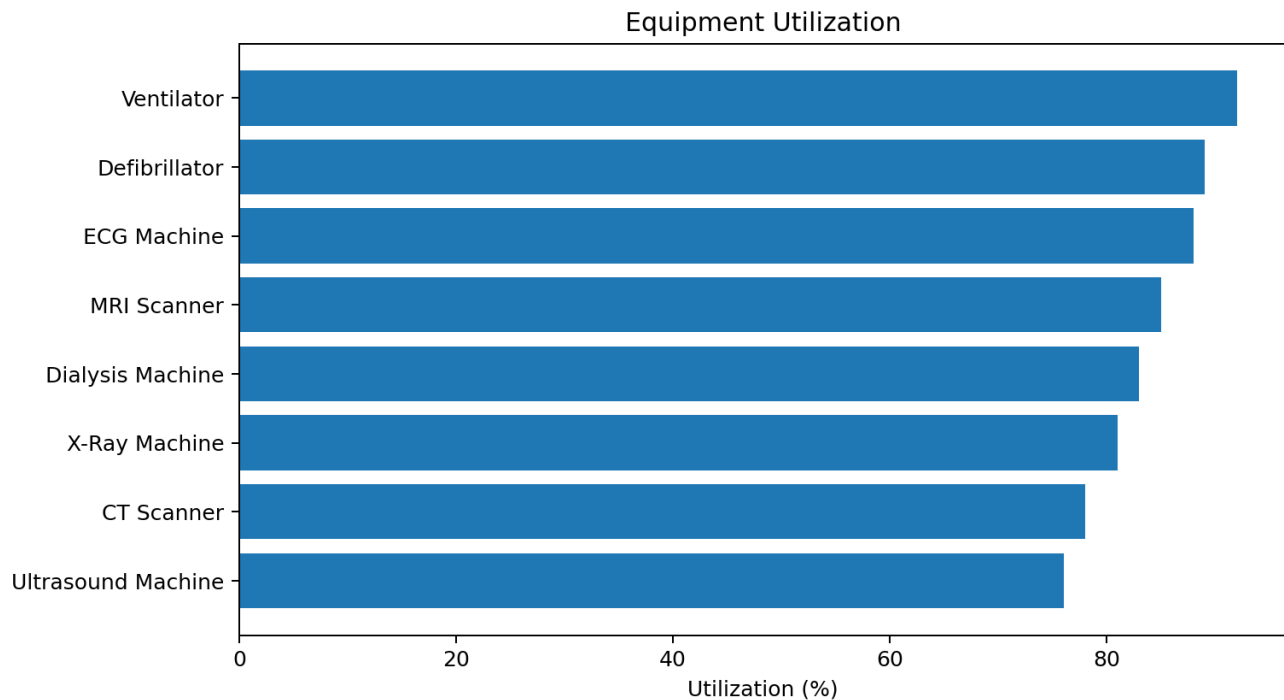


Figure 23. Equipment utilization reference.

Reference note: Equipment utilization should be refreshed from the approved workbook rather than edited manually in the dashboard. Any change in equipment naming or department mapping should be reflected in the source workbook and QA process.

Appendix D – Source Traceability Matrix

Evidence Item	Used For	Traceability Statement
Project specification	Requirements and Module 8 criteria	Defines the six KPIs, four dashboards and documentation deliverables.
Hospital CSV	Operational metrics	Primary analytical source for admissions, LOS, departments, cost, beds and staffing.
Equipment workbook	Equipment utilization	Provides equipment, department and utilization percentage.
KPI Python script	Calculation methodology	Defines validation, cleaning, KPI formulas and output files.
Dashboard PDF	Visual implementation evidence	Shows the four submitted dashboard pages and headline KPI values.

This matrix is deliberately conservative: it identifies what the supplied files support and avoids adding unsupported external claims.

Submission readiness

Before external publication, add the confirmed original dataset provider and URL to this matrix and to the repository README if that information is available from the project owner.

Traceability note: Source traceability is a quality attribute. A reviewer should be able to identify which file supports each major statement, metric or visual. When an external dataset URL becomes available, it should be added to the repository and this matrix.

Appendix E – Professional Review and Sign-off

Review Area	Reviewer Check	Sign-off
Data	Source, row count, fields and privacy reviewed	_____
KPI	Definitions and formulas reviewed	_____
Dashboard	Four pages, filters and labels reviewed	_____
Healthcare methodology	Operational interpretation reviewed	_____
QA	Validation evidence reviewed	_____
Documentation	Formatting, completeness and references reviewed	_____
Delivery	Repository structure and final files reviewed	_____

Prepared for Module 8 – Documentation and Project Delivery.

Project: MedTrack_DV – Hospital Operations & Patient Analytics Dashboard

Document version: 1.0 | Date: 17 August 2026

Final quality statement

This document is designed as a complete project handover artifact: requirements, evidence, methodology, dashboard usage, data interpretation, governance and future improvements are consolidated into one controlled report.

Release-control summary

Data integrity. Confirm the final dataset version, row count, required columns, date range and resource files before packaging.

Metric integrity. Confirm the six KPI definitions, calculation assumptions and dashboard headline values against the approved QA snapshot.

Visual integrity. Confirm that every dashboard page has readable titles, units, filters and labels and that screenshots correspond to the submitted dashboard version.

Documentation integrity. Confirm that the project structure, methodology, limitations, source traceability and future roadmap are included in the final repository.

Release Item	Required Evidence	Final Check
Source files	CSV + equipment workbook	Verified
Calculation	KPI script + outputs	Verified
Visualization	Four dashboard views	Verified
Documentation	Module 8 report	Verified